

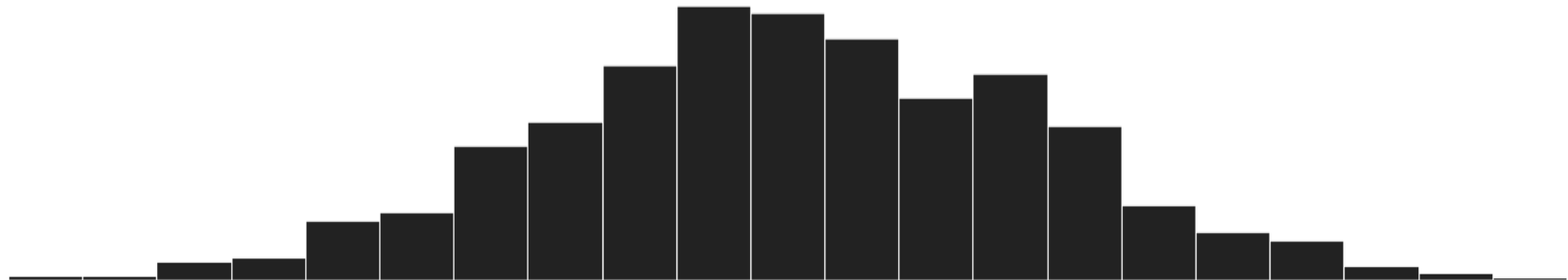
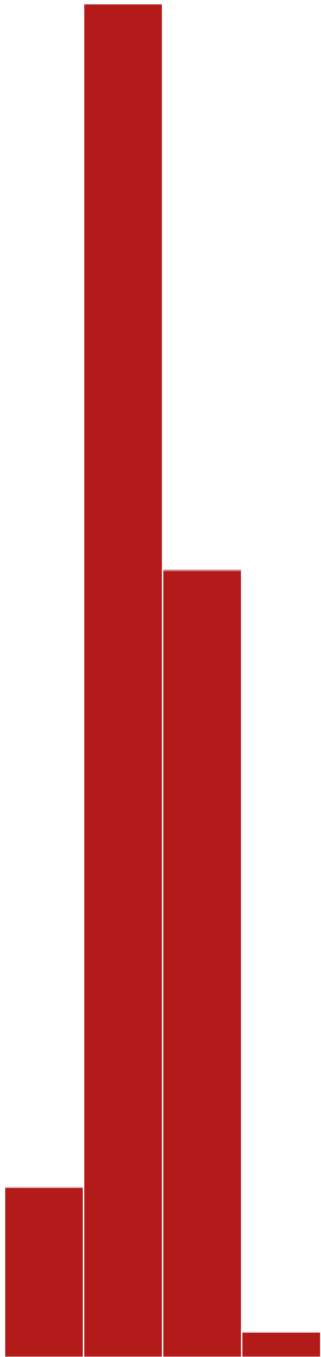
VTPEH 6105

Lecture 03

INFERENCE & HYPOTHESIS TESTING

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About me and the org

I'm Edward, but many of you probably know me by my nickname Teddy. I am an attorney and the co-founder of People and Earth Law Group, a 501(c)(3) non-profit dedicated to providing services related to animal/environmental law. Want to get in touch? Depending on your preference, you can either reach out via email, phone, or through our website.

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peopleandearthlaw.org



How to Make the Most of this Class?

- In class
 - Don't need to code everything, the code is available on Canvas
- After class
 - Do the Knowledge Checks
 - Focused on basic concepts and skills needed for the rest of the course
 - Think about the answers, retake as needed
 - Do the Guiding Quiz if needed
 - Think about how it connects to the Homework (it should almost give you the answers...)
 - **Come with questions**
 - To VTPEH 6108 (post on Canvas)
 - To office hours
- About the Homework
 - **Read the comments** on Canvas – CAs spend a lot of time giving you constructive feedback

How to Make the Most of this Class?

- Office hours
 - **Tuesdays, 09:00-10:00 AM** (Jordan, Maddy)
 - **Wednesdays, 05:00-06:00 PM** (Mainul, Princal, Daniel and Harsh)
 - If you cannot make it at these times, please contact the course assistant of your choice by email
 - In S1-122, unless indicated otherwise on Canvas
- Any topic you would like to chat about? [VTPEH 6108 Topic Suggestion](#)
Drop a message on the dedicated discussion (anonymous). We'll do our best to accommodate
- Struggling with coding? [DataCamp Classroom](#) (and other resources on Canvas)
Free. Online. Hands-on. Self-paced. Interactive.

How to Make the Most of this Class?



Account

Dashboard

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VTPEH6105 > Modules

Spring 2026

Home

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Recent Announcements



Today's class (01/30, LH3)

Hi everyone! Today, we will meet in LH3 again. Please download all the material (available on the Module page) before entering the room, as WiFi isn't the b...

Reply

Posted on:

Jan 30, 2026, 8:31 AM

Collapse All

Resources

VTPEH - Getting Started with R.pdf

VTPEH 6108 Topic Suggestion

DataCamp Classroom (free online R courses)

View Course Stream

View Course Calendar

View Course Notifications

To Do

Guiding Quiz #01

VTPEH 6105
Biostatistics for Health Sciences (2026SP)
15 points |
Jan 29 at 5pm

Homework #01

VTPEH 6105
Biostatistics for Health Sciences (2026SP)
16 points |
Jan 29 at 5pm

Knowledge Check #01

VTPEH 6105
Biostatistics for Health Sciences (2026SP)
10 points |
Jan 30 at 11am

How to Make the Most of this Class?

About the use of AI in particular...

If you use AI to generate code you don't understand...

1. Check that the output make sense

Subsetting data? Should be smaller than the original data set, not full of NAs...

Expecting a probability? Should be between 0 and 1

2. Decompose the code to understand what each command does

- Make sure you understand what each symbol does (calling a function, calling a position in a vector, calling a vector in a data frame, testing a condition...)
- Make sure you understand what each function does
- Run each command you don't know one by one (including those hidden in nested functions) to see (and understand) what it does

How to Make the Most of this Class?

About homework in particular...

- Review the PDF before submitting. Ensure this is shaped like a document you would share with collaborators. Develop good habits now!
- Include all the required file (.pdf *and* .rmd, so we can check where the issue comes from if there's any). More broadly, read the instructions!
- If you submit late, I grade and you get my rushed comments instead of the thoughtful, constructive comments from the CAs!

1 Instructions

This homework must be submitted on Canvas as a **R markdown-generated PDF** file (from the .Rmd template available on the Assignment page), **together with the source code** (the .rmd file). Submissions in any other format will be considered late until the requested format has been submitted; failure to submit the requested format within 10 days will lead to a 0 to the assignment.

Questions should be answered with **literate, meaningful answers** when relevant.

Feedback on Previous Homework

- Using informative statistics...
 - Interquartile range ($Q_3 - Q_1$) vs interval ($[Q_1; Q_3]$)
Which one is the most informative? Think about dispersion but also symmetry...
- Reporting results
 - Actually report the numbers
 - Round them appropriately
 - Include units and other useful information
 - No need to say “about”

The... I chose...

The mean age is 11.01, and the ~~standard error~~ is 5.033.

The mean participant age was 11.0 years with a standard deviation of 5.0 years.

The mean age is 11.01 years (SD = 5.03).

Feedback on Previous Homework

- Common barriers to knitting R markdown documents...

Read the error message

- The PDF is open and can't be overwritten
- There's a line of code prompting an action from the coder
E.g., `install.packages()` for an already installed package
- There's a dependency that can't be found within the R markdown script
E.g., a function from a package that isn't been loaded with `library()`
E.g., a data set that isn't located in the same folder as the R markdown script
- There's non-R code within R chunks
- There're several chunks with the same name
- There's a bug in the code (there should never be any bug in your code!)

Feedback on Previous Homework

- Common barriers to knitting R markdown documents...

Chunk name

R code chunk

```
19 {r setup, include = FALSE}
20 knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)
21
22
23
24
25 # Introduction
26
27 We will use the [CDC Malaria
data](https://data.cdc.gov/Global-Health/Number-of-Reported-Malaria-Cases-by-County-United-/puzh-
5wax/about_data), and the [CDC Social vulnerability Index (SVI)
data](https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html).
28
29 {r|, results='hide'}
30
31 data_malaria = read.csv("Data/Number_of_Reported_Malaria_Cases_by_County__United_States__2017_202
50120.csv")
32 data_svi = read.csv("Data/SVI_2016_US_county.csv")
33
34
```

Run that chunk (should run!)

Run all the chunks above (should run!)

- There's non-R code within R chunks
- There're several chunks with the same name
- There's a bug in the code (there should never be any bug in your code!)

Feedback from the Class

- Struggling with anything? Talk to me or to the CAs!
- Adjustments to the Assignments
 - No assignment will be re-graded
 - For the Homework: possibility to drop one from your final grade
 - Reach out at the end of the semester, and let us know if and which Homework # you would like to drop
 - Only Homework assignments are eligible (not the quizzes – those you can take as much as you want/need)
 - Opportunity for bonus points: develop your own R command glossary / cheat sheet
 - Optional (not mandatory), graded as bonus
 - Can be any format (R markdown html page, Illustrator poster, hand-written card...)
 - Should include commands introduced in at least 2 different classes (i.e., is updated as you learn)
 - Assignment and examples will be posted this week
 - Will be due at the end of the semester, but can be submitted earlier
 - Main goal: have you reflect on what you have learned, and develop a useful resource
 - In particular: great way to reflect on AI-suggested commands

First Steps into Statistical Inference

Point and Interval Estimates (e.g., Confidence Intervals)

Now sit comfortably and relax...

Why Do We Do Statistics?

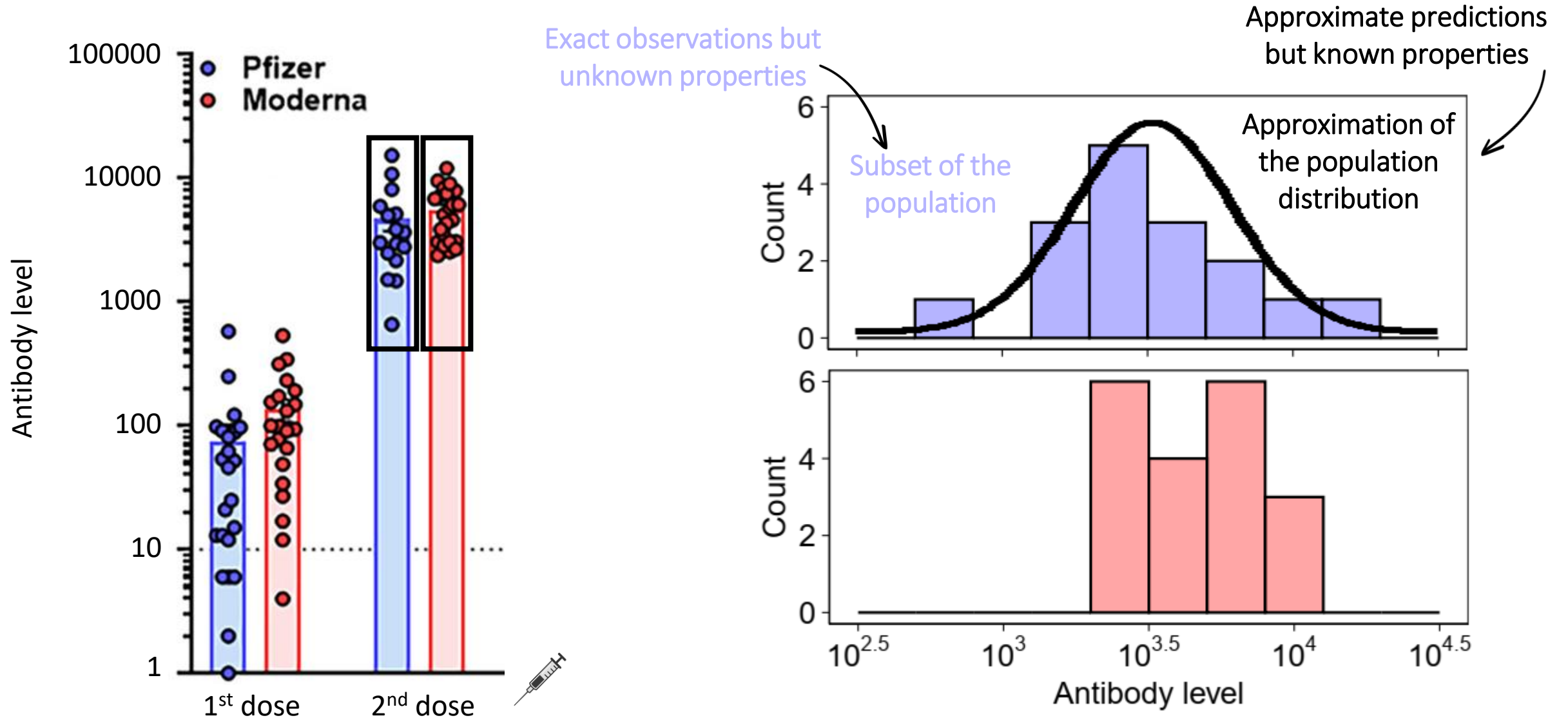
- Descriptive statistics
- Statistical inference

How Do We Do Statistics?

- We draw on **probabilities**
- **Probability**
Given known parameters, what outcomes can we expect?
- **Statistics**
Given observed outcomes, what can we infer about unknown parameters?

Inference = data-informed guess

Probability Distributions: Why Do We Care?



Probability Distributions: Why Do We Care?

Probability distribution

Mathematical function that gives the probabilities of occurrence of different possible outcomes for an experiment



Uncertainty

Prediction is limited to probabilities

Approximates and (kinda) predicts the data

Universal question

- How to interpolate missing data?
- How to predict trends?

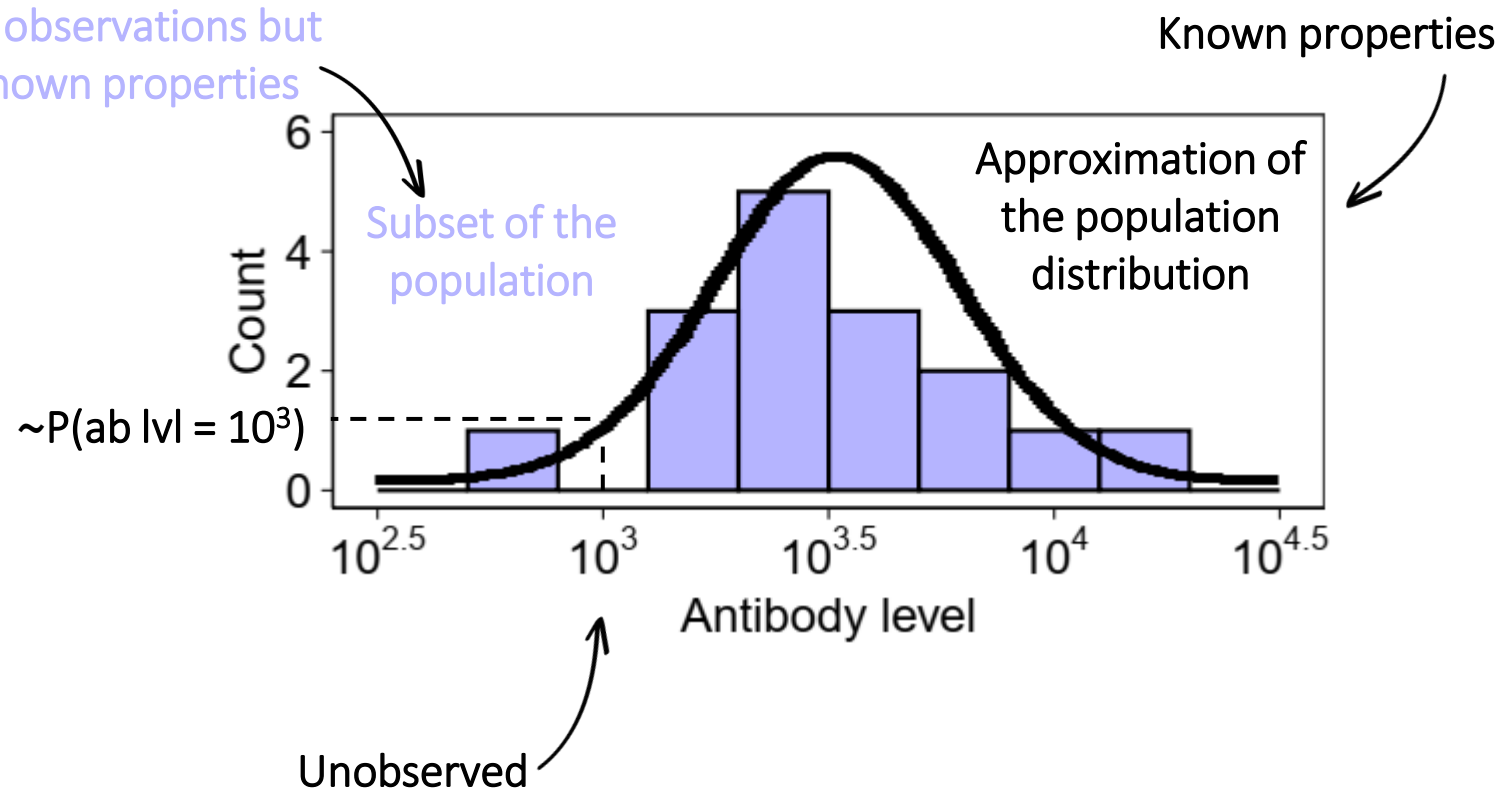
Frequency Distribution vs Probability Distribution

- **Frequency**

A measure of how often events occur, i.e., the **exact number of times each value occurs**

- **Probability**

A measure of how likely events are to occur, i.e., the **probability of occurrence of each value**

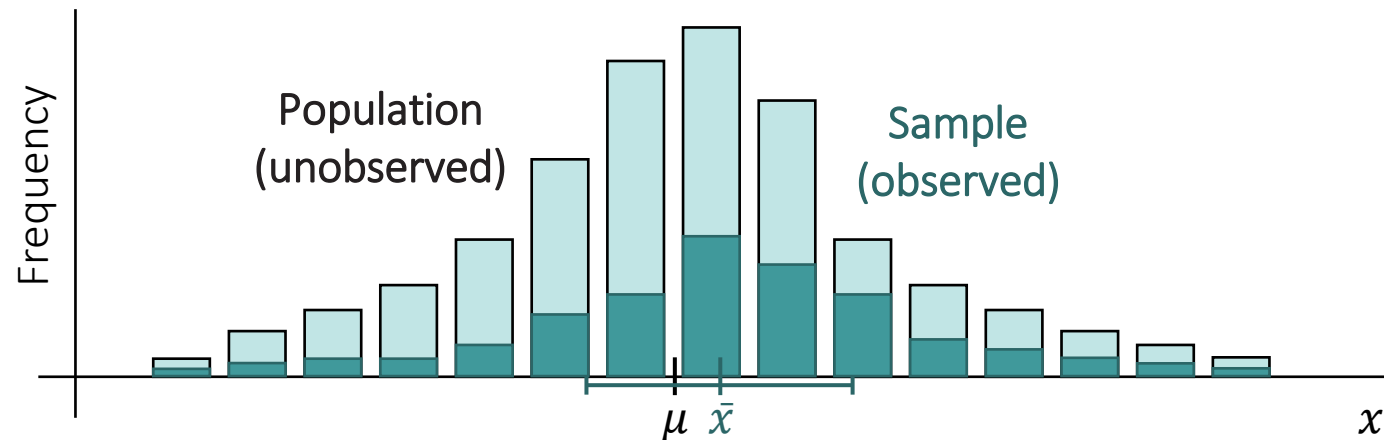


Estimates

- In statistics, **estimation** refers to the process by which one makes **inferences** about a population, based on information obtained from a sample

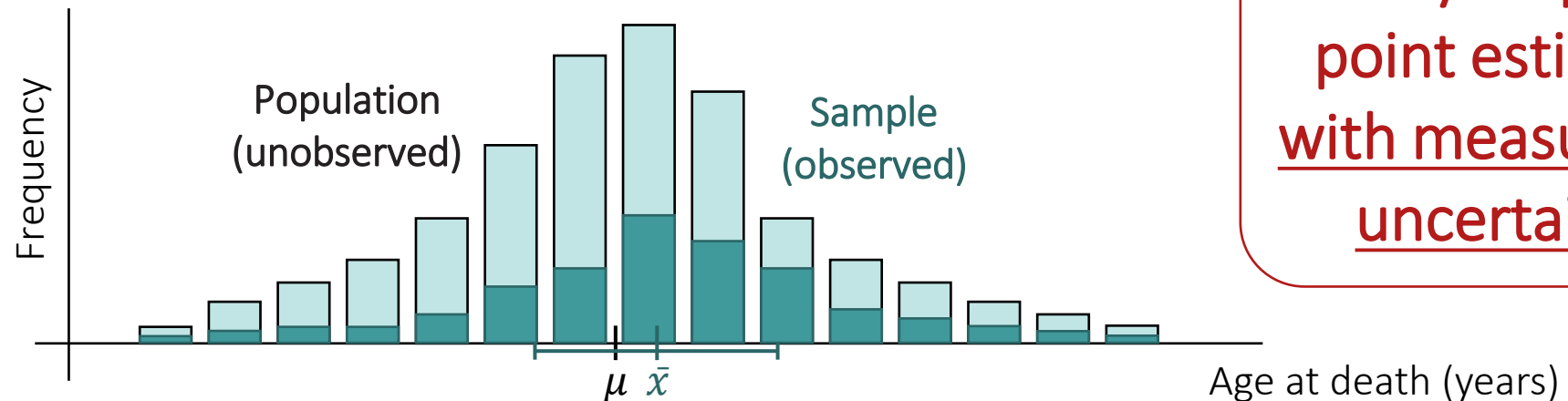
We **estimate parameters** based the **observed data**

- **Point estimation** – using the sample data to calculate a single number to estimate the parameter of interest
E.g. use the sample mean $\bar{x}$ to estimate the population mean μ (but we do not know how close the $\bar{x}$ is to μ)
- **Interval estimation** – calculate a range of reasonable values that are intended to contain the parameter of interest with a certain degree of confidence



Estimates

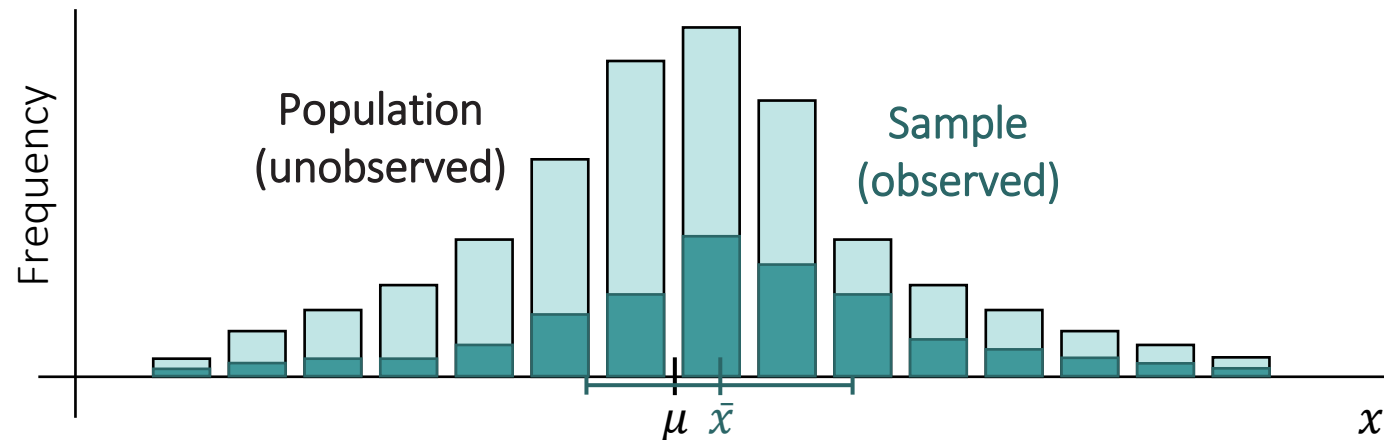
- To describe a sample, always report a **central tendency** measure with a **dispersion** measure
E.g., in the sample, average ($\pm$ standard deviation) age at death is 78.3 ± 4.5 years
Or, in the sample, median [1^{st} – 2^{nd} quartiles] age at death is 77.1 [68.2 – 83.9] years
- When inferring population parameters, always report a **point estimate** with a measure of uncertainty **as an interval**
E.g., in the population, average [95% confidence interval] age at death is 78.3 [69.1 – 87.5] years



**Always report a
point estimate
with measures of
uncertainty**

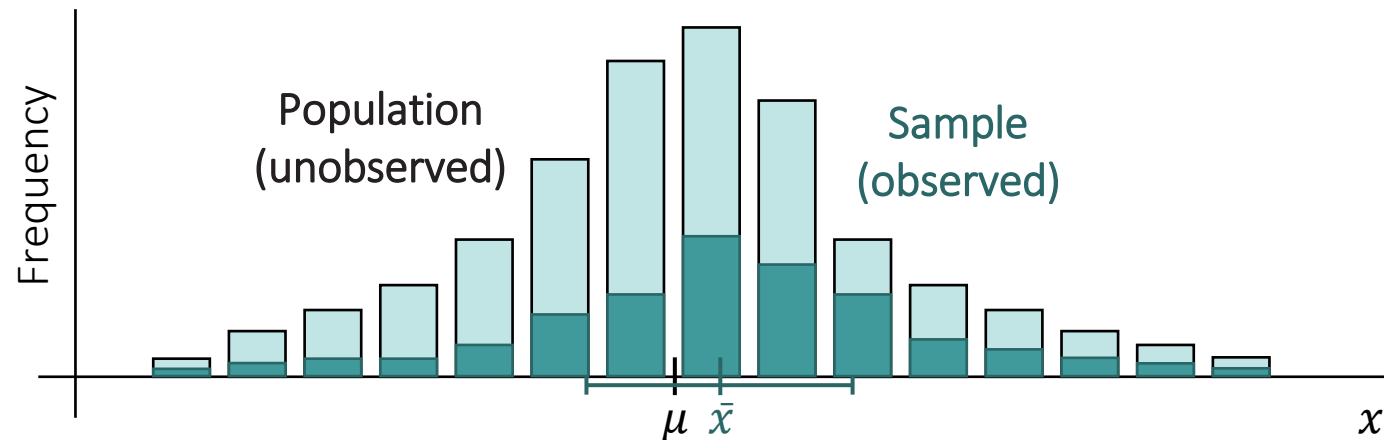
Estimates

- Spoiler: getting confidence intervals is 1 line of code in R or Excel...
- Disclaimer: this is not about learning how to calculate confidence intervals (you can ask AI), this is about understanding the concepts behind to correctly interpret and/or use them



Estimates

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$... and thanks to the central limit theorem, we can infer μ



Behind the Scene: Quantifying (Un)certainty

Thanks to the Central Limit Theorem and the Normal Distribution

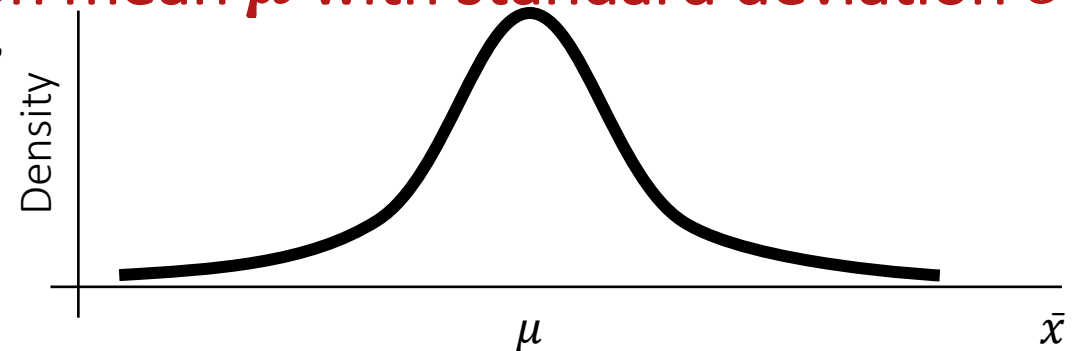
The Central Limit Theorem to the Rescue

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$
- **Central limit theorem**: given a population of any (non-normal) functional form with a mean μ and standard deviation σ , the sampling distribution of $\bar{x}$, computed from samples of size n from this population, will have mean μ and standard deviation $\sigma/\sqrt{n}$ and will be approximately **normally distributed** when the sample size is large

$$\bar{x} \sim N(\mu, \sigma/\sqrt{n})$$

→ **For n large enough, the distribution of the sample mean $\bar{x}$ is approximately normal, and centered around the population mean μ with standard deviation σ**

- Rule of thumb: $n \geq 30$ considered “large”
- Side note: variance = standard deviation²



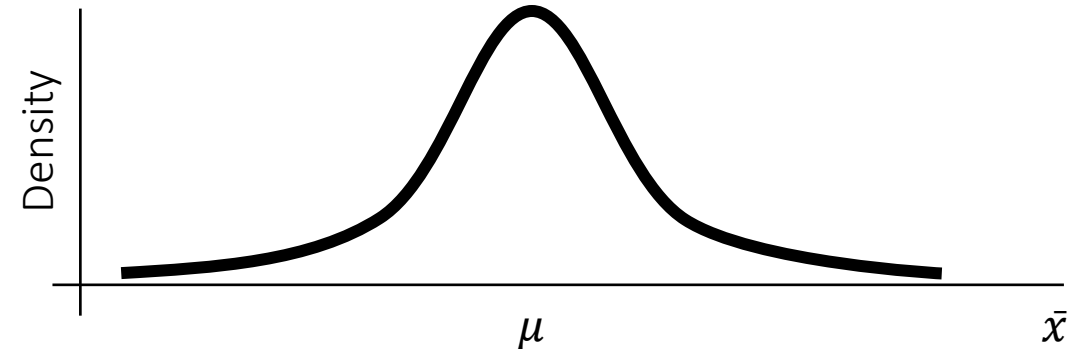
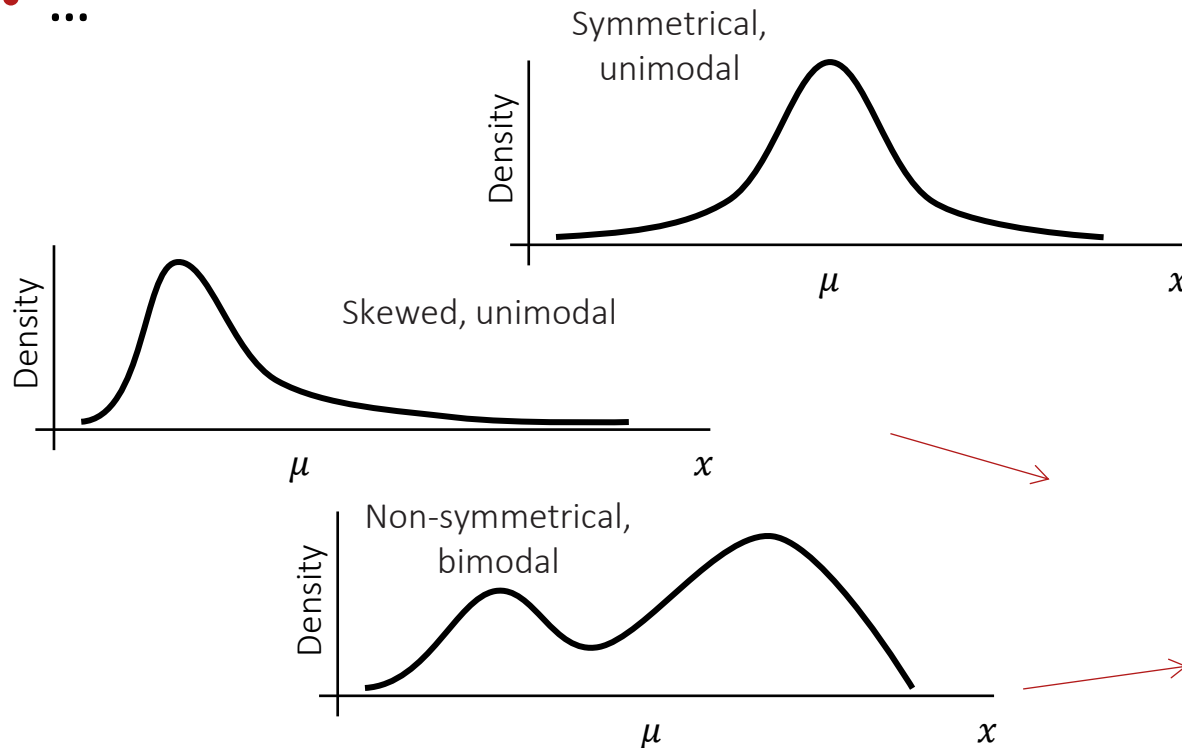
The Central Limit Theorem to the Rescue

- Population of 100,000 individuals with mean μ

- Sample #1 of 5,000 individuals with mean $\bar{x}_1$
- Sample #2 of 5,000 individuals with mean $\bar{x}_2$
- Sample #3 of 5,000 individuals with mean $\bar{x}_3$
- Sample # i of 5,000 individuals with mean $\bar{x}_i$
- ...

$\bar{x}$ follows a normal distribution centered around μ with standard deviation $\sigma/\sqrt{n}$

Independently from the distribution of x



The Central Limit Theorem to the Rescue

- If we know the sample mean, we can infer the population mean...
- ... Because the sample mean follows a normal distribution
- And why is this cool? Because we know everything about the normal distribution!

The Normal Distribution

- Use to describe continuous variables with symmetric distributions
 - Examples?
 - Size, weight... lots of biological variables

Normal distribution

$$P(x) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Parameters

μ : mean

σ : standard deviation

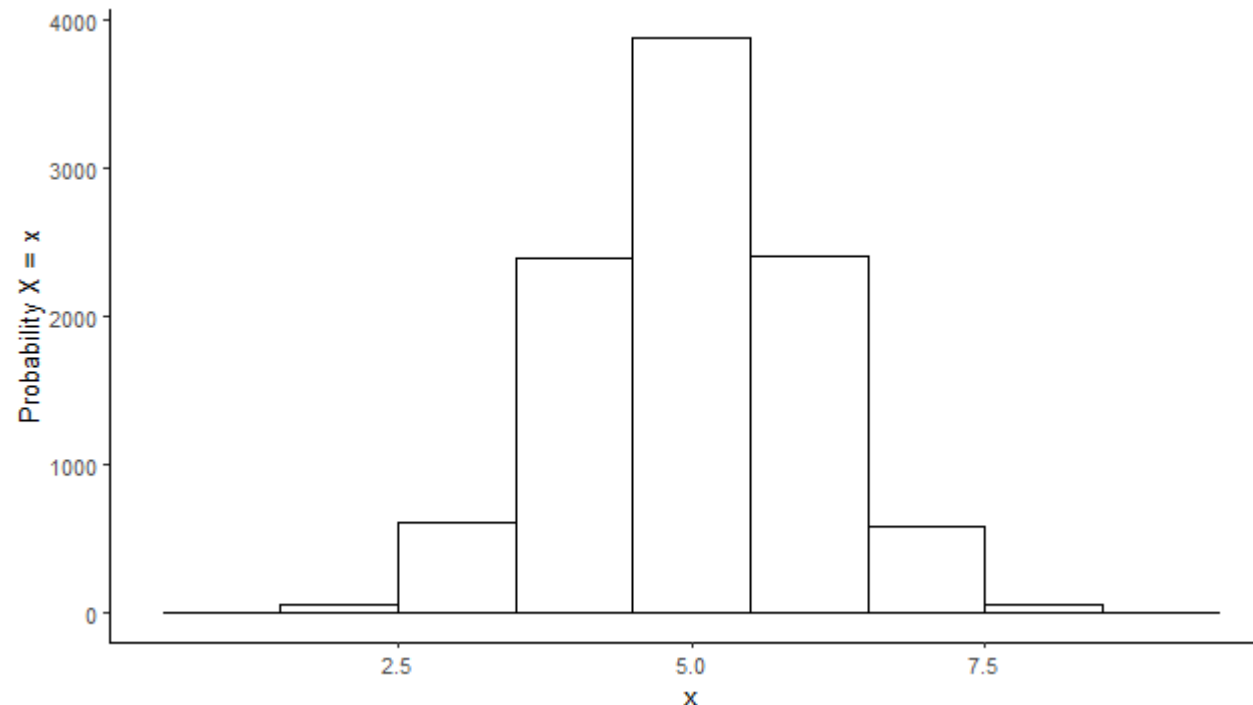
* You do not need to know this, you just need to know we (collectively as humans) know it, so we can do math with it

The Normal Distribution

- Probability density function (PDF)

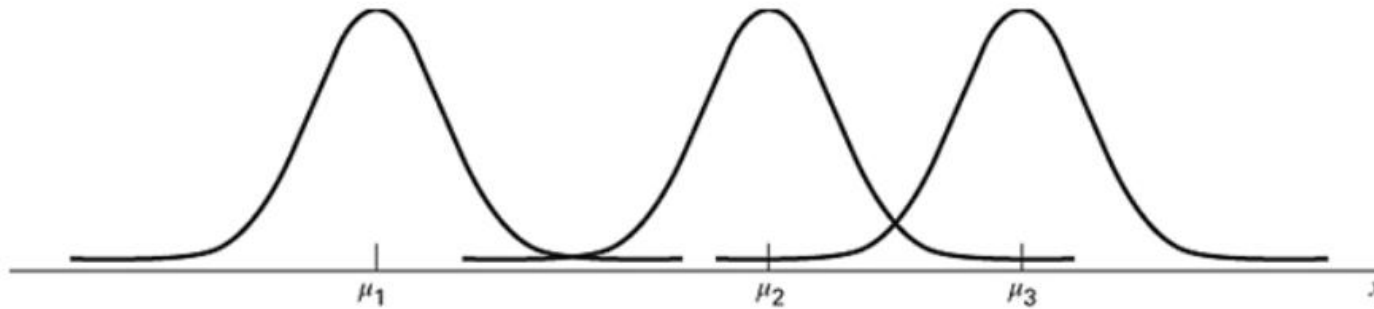
$$f(X) = P(X = x) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- X is a continuous random variable
 - Expected value $E[X] = \mu$
- $f(X)$ is a probability $f(X) \in [0; 1]$
Variance $\text{Var}[X] = \sigma^2$

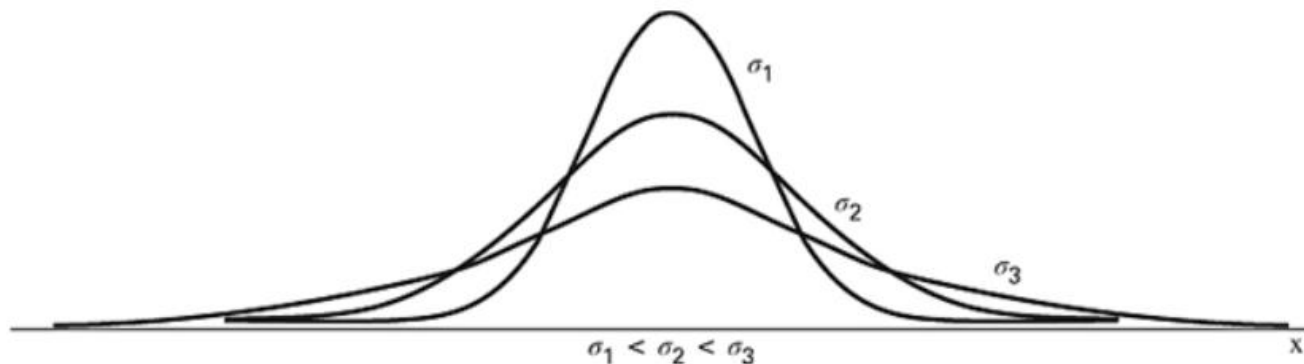


The Normal Distribution

- Use to describe continuous variables with symmetric distributions
 - The mean and variance completely describe the normal curve



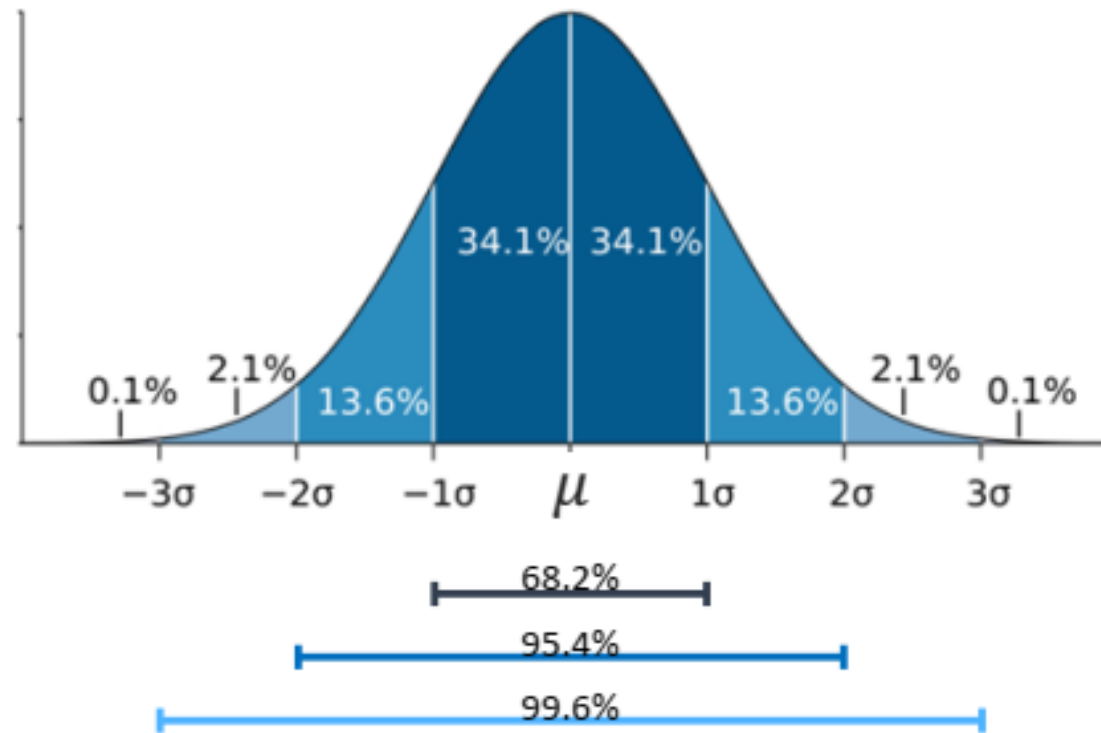
$$\mu_1 < \mu_2 < \mu_3$$



$$\sigma_1 < \sigma_2 < \sigma_3$$

The Normal Distribution

- Use to describe continuous variables with symmetric distributions
 - The mean and variance completely describe the normal curve



The Normal Distribution

- The curve is called a probability density function or the bell-shaped curve.
- Continuous random variable
- Symmetric around its mean, μ
- The mean = median = mode (unimodal)
- The total area beneath the curve sums to 1
- σ the standard deviation = the amount of dispersion around the mean

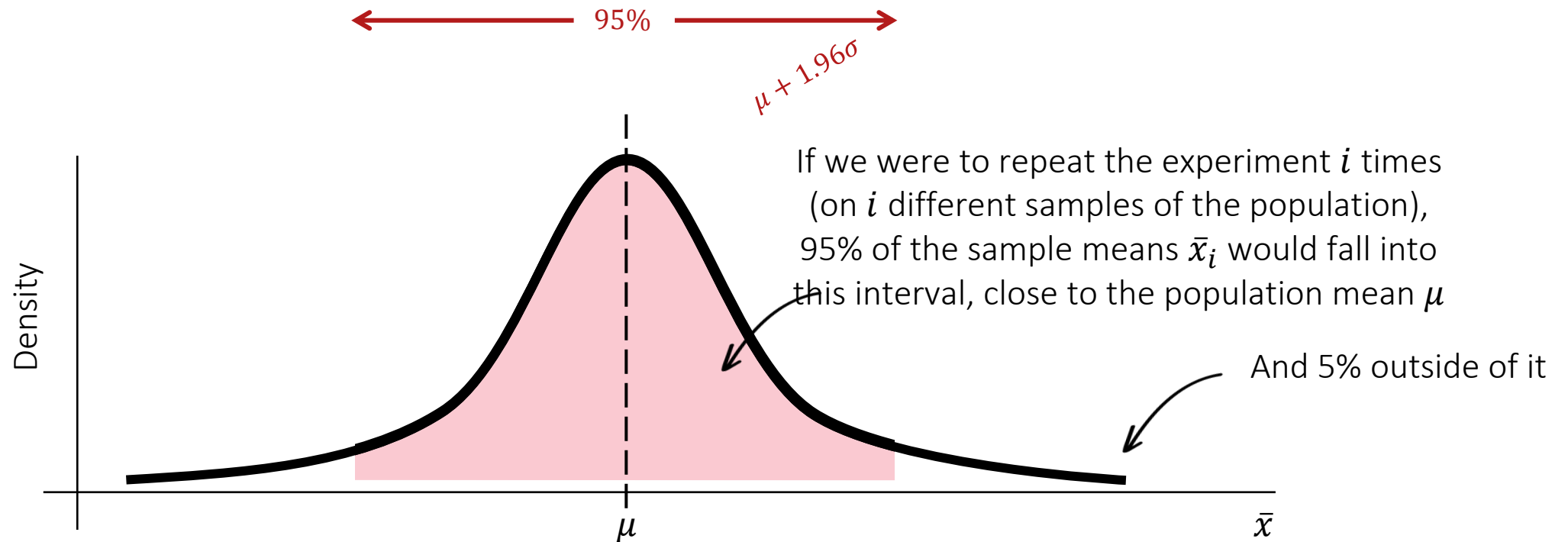
The Central Limit Theorem to the Rescue

- If we know the sample mean, we can infer the population mean...
- ... Because the sample mean follows a normal distribution
- And why is this cool? Because we know everything about the normal distribution!
- It does not mean we know exactly where the true population mean is...
- It means **we know where the true population mean likely is, $\pm$ some uncertainty...**
- ... Which we can quantify from what we know about the normal distribution

We never know the true population mean, but we know the distribution of the sample mean is centered around it, so we can infer where it likely is

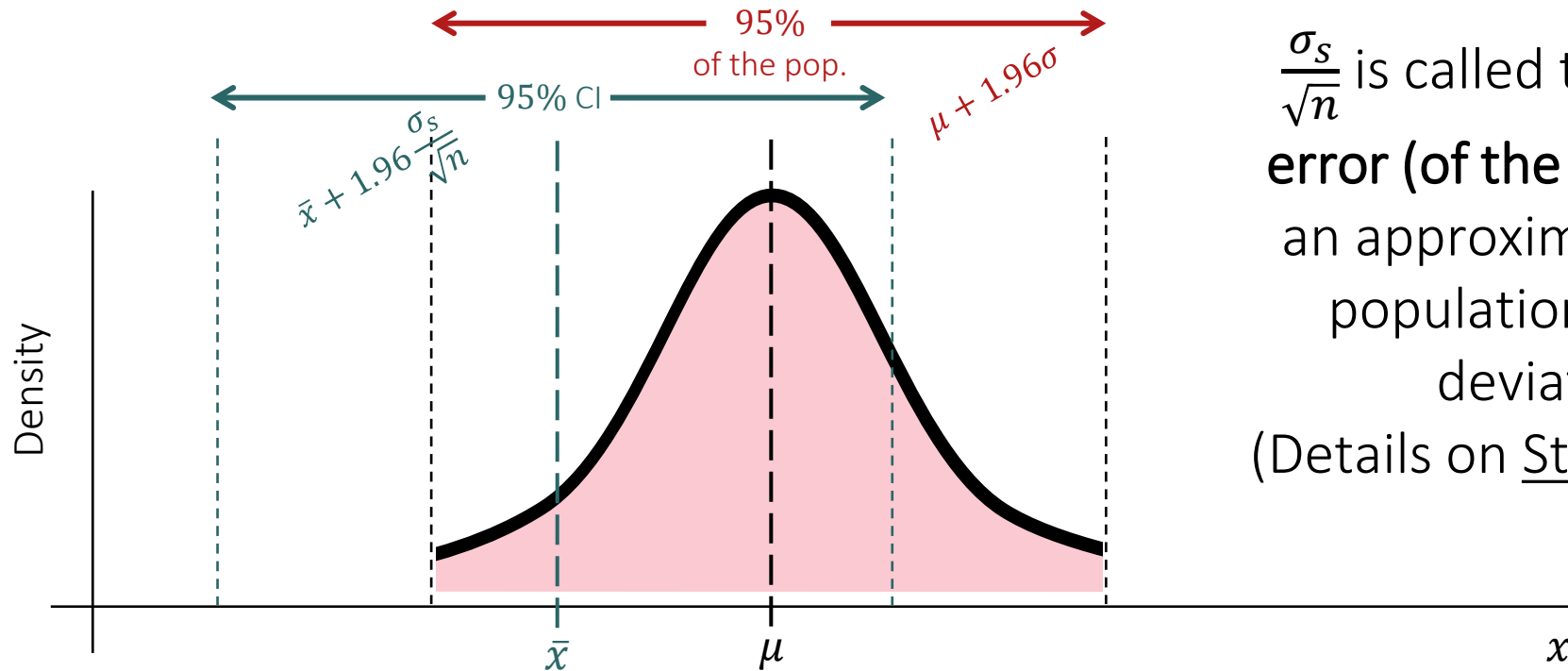
Confidence Intervals: Definition

- If we knew the population parameters...



Confidence Intervals: Definition

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$

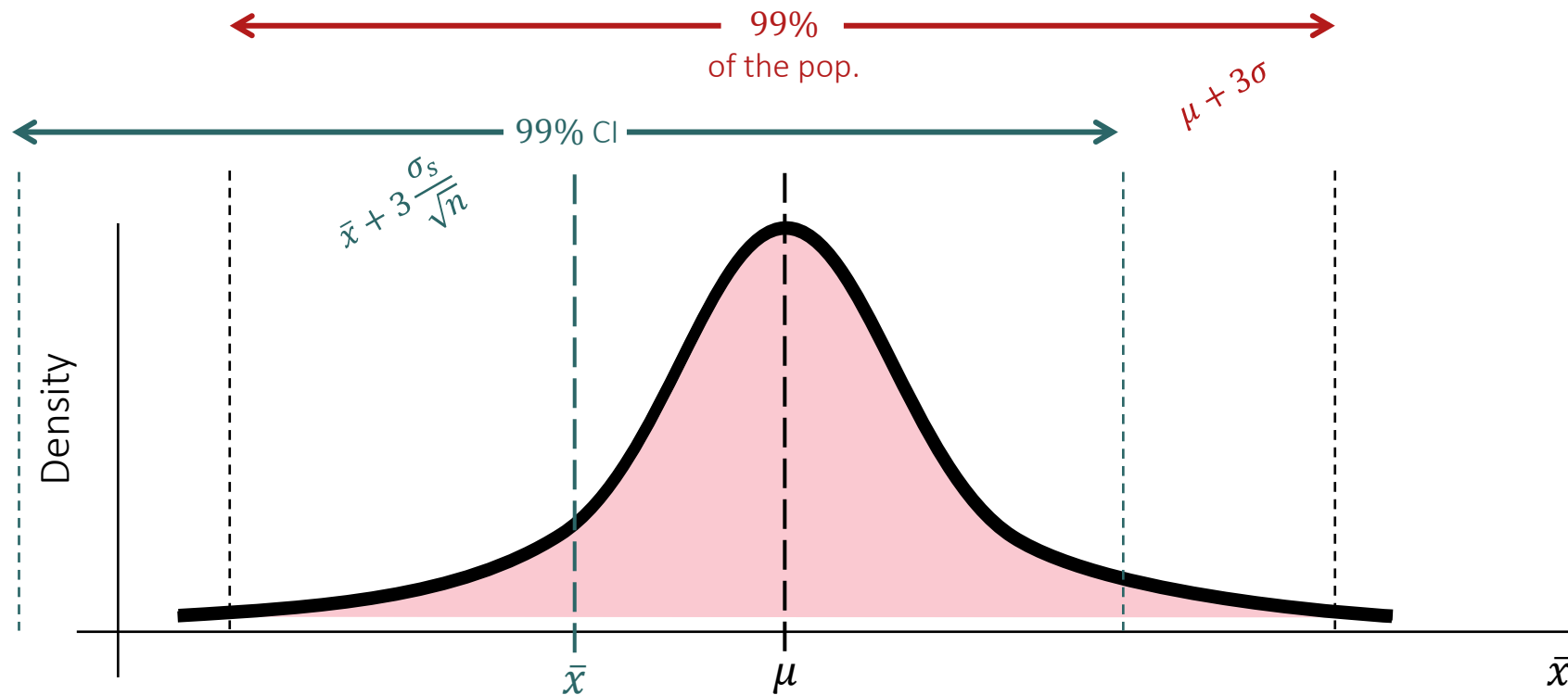


$\frac{\sigma_s}{\sqrt{n}}$ is called the **standard error (of the mean)** and is an approximation of the population standard deviation σ
(Details on [StackExchange](#))

95% CI = $[\bar{x} - 1.96 \frac{\sigma_s}{\sqrt{n}}; \bar{x} + 1.96 \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, and n the sample size

Confidence Intervals: Definition

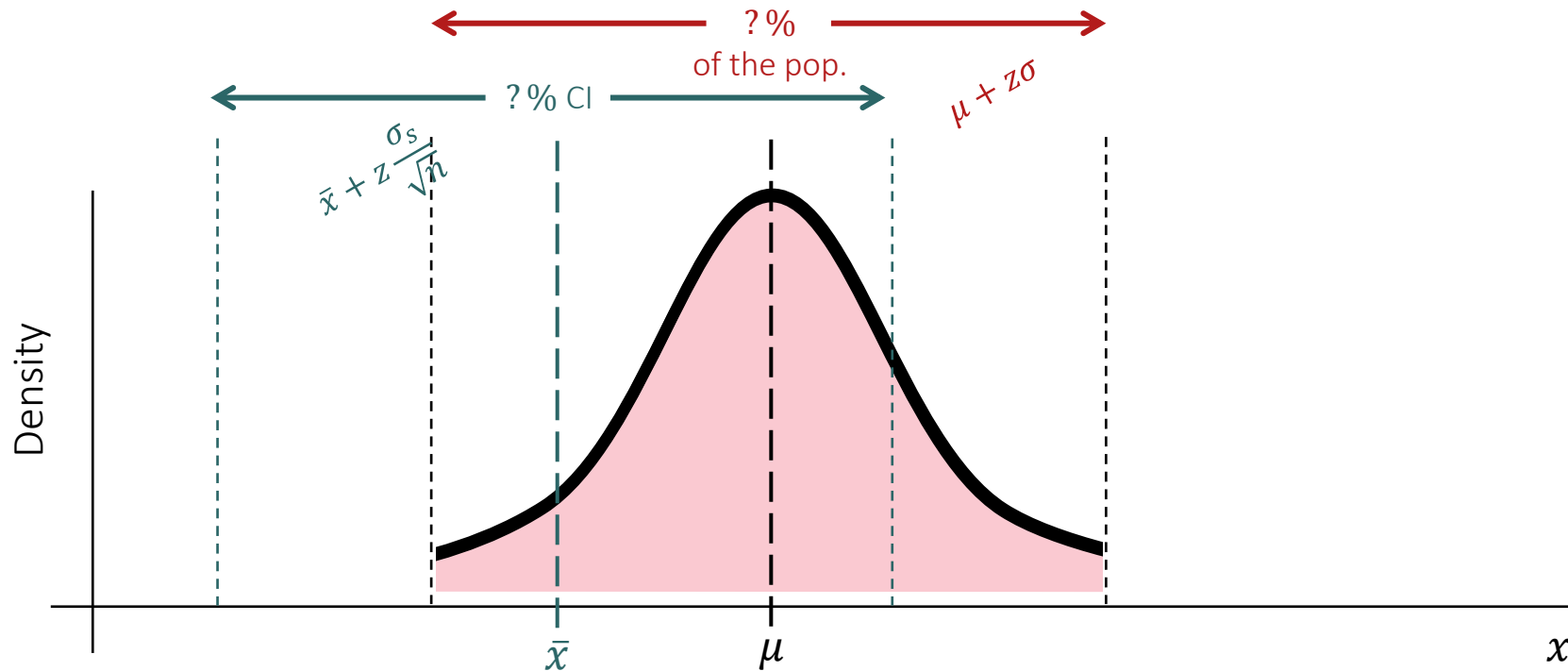
- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$



99% CI = $[\bar{x} - 3 \frac{\sigma_s}{\sqrt{n}}; \bar{x} + 3 \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, and n the sample size

Confidence Intervals: Definition

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$

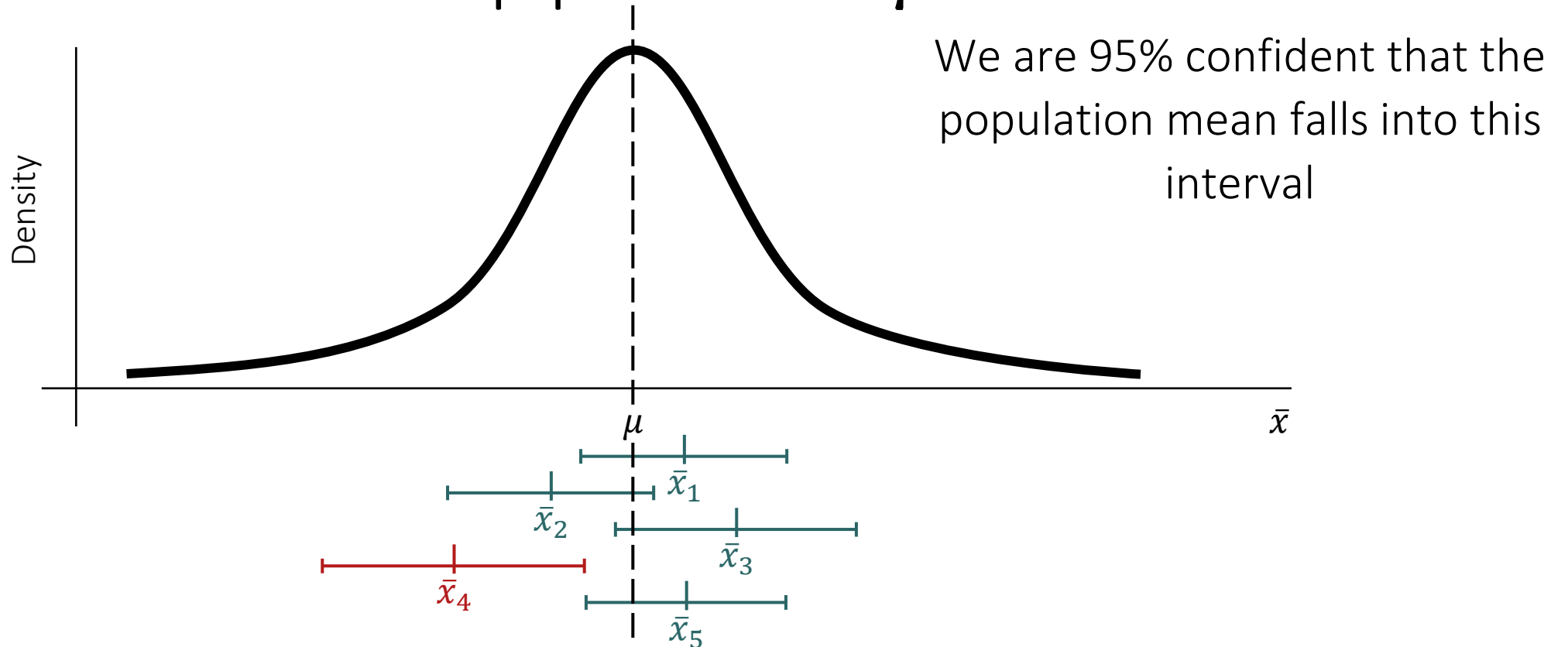


$CI = [\bar{x} - z \frac{\sigma_s}{\sqrt{n}}; \bar{x} + z \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, n the sample size and z the z-score for the confidence level

Confidence Intervals: Interpretation

- Starting from the sample parameters...

For 95% confidence intervals, if we were to repeat the experiment i times (on i different samples of the population), **95% of the i defined confidence intervals would contain the true population mean μ**



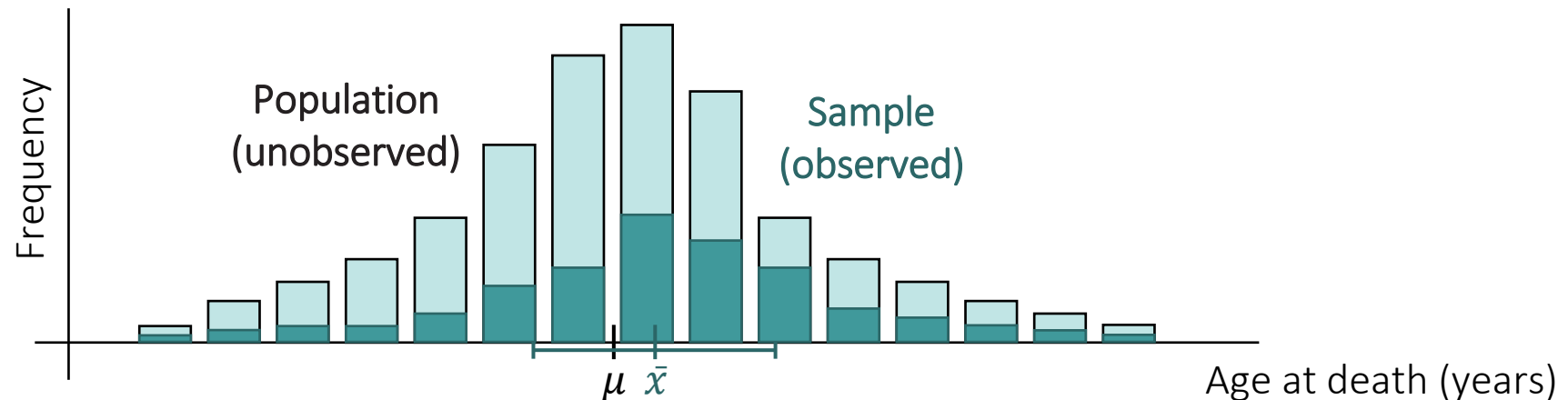
Confidence Intervals: Interpretation

- To describe a sample, always report a **central tendency** measure with a **dispersion** measure

E.g., in the sample, average ($\pm$ standard deviation) age at death is 78.3 ± 4.5 years. When inferring population parameters, always report a **point estimate** with a measure of uncertainty **as an interval**

E.g., in the population, average [95% confidence interval] age at death is $78.3 [69.1 - 87.5]$ years

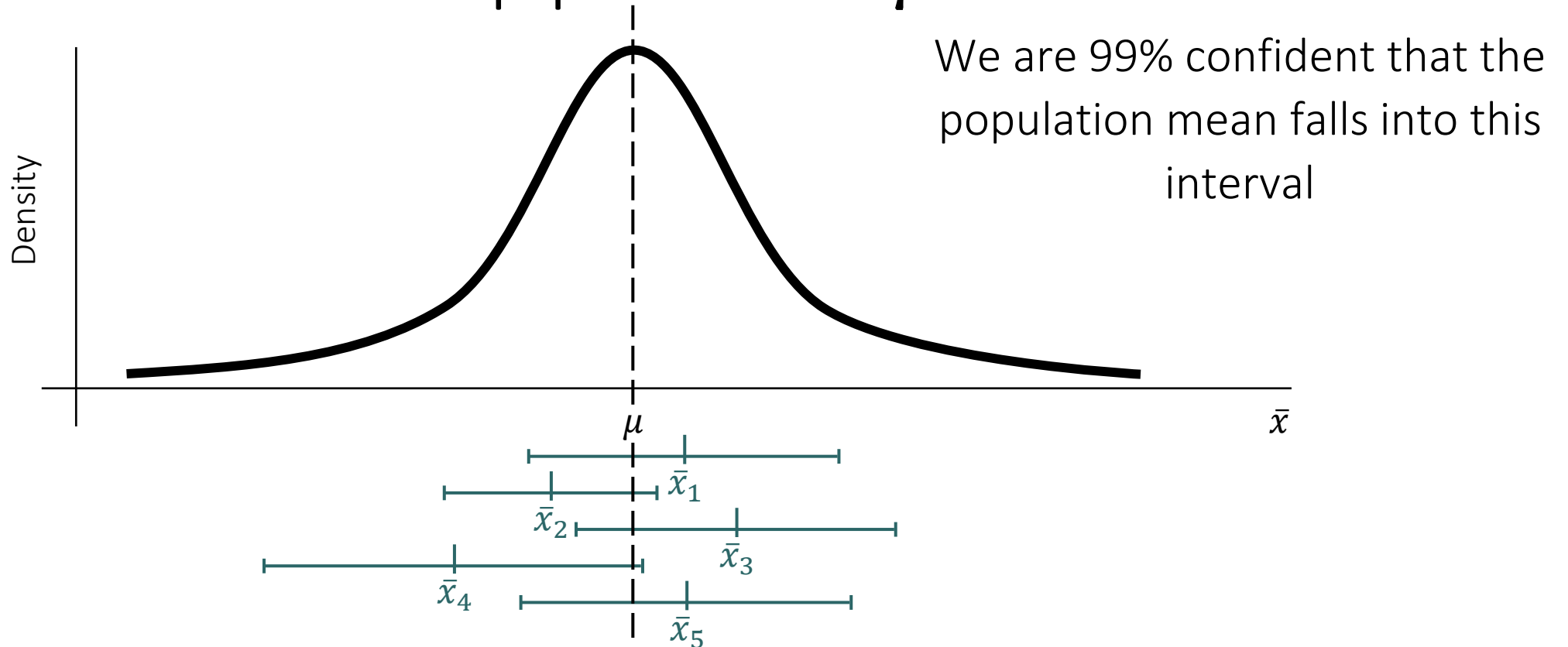
→ We are 95% confident that the average age at death in the population falls into the interval $78.3 [69.1 - 87.5]$ years



Confidence Intervals: Interpretation

- Starting from the sample parameters...

For 99% confidence intervals, if we were to repeat the experiment i times (on i different samples of the population), **99% of the i defined confidence intervals would contain the true population mean μ**



Confidence Intervals: Interpretation

- Starting from the sample parameters...

For 99% confidence intervals, if we were to repeat the experiment i times (on i different samples of the population), **99% of the i defined confidence intervals would contain the true population mean μ**

- The **wider** the confidence interval, the **more confident** we are to overlap with the population parameter, but the **less informative** the estimated parameter is
- The higher the sample size n , the tighter the confidence interval

Increasing sample size enables a higher precision with high confidence
(i.e., lower uncertainty)

Confidence Intervals: Interpretation

- The same procedure applies to means and proportions

Mean

$CI = [\bar{x} - z \frac{\sigma_s}{\sqrt{n}}; \bar{x} + z \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, n the sample size and z the z-score for the confidence level

E.g., age at death in a population

Proportion

$CI = [\bar{p} - z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}; \bar{p} + z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}]$
with $\bar{p}$ the sample proportion, n the sample size and z the z-score for the confidence level

E.g., number of infected people in a population (prevalence)

Confidence Intervals: Why Do We Care?

- Reporting point estimates with confidence intervals provides information on the uncertainty around the point estimates

It is a critical information to interpret numbers

Never trust a mean not presented with its confidence interval!

- It is the first step in **hypothesis testing**

Most hypothesis testing procedures are based on comparisons of means

Building Intuitions around Probabilities

The Binomial Distribution

The Binomial Distribution

- Used to describe binary events, by counting the occurrence of one event or the other
 - Examples?
 - Coin flip
 - The number of heads we obtain (**x**) is based on the probability of flipping a head (**p**) and the number of times we toss the coin (**n**)

$$P(x \text{ heads}) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- $P(2 \text{ heads})$ when tossing the coin 3 times
 - $x = 2; n = 3$
 - $p = 0.5$
 - The probability of getting 2 heads among 3 tosses =

probability of getting 2 heads AND 3-1 = 1 tails

in every possible order (combinations without replacement)

$$= 0.375$$

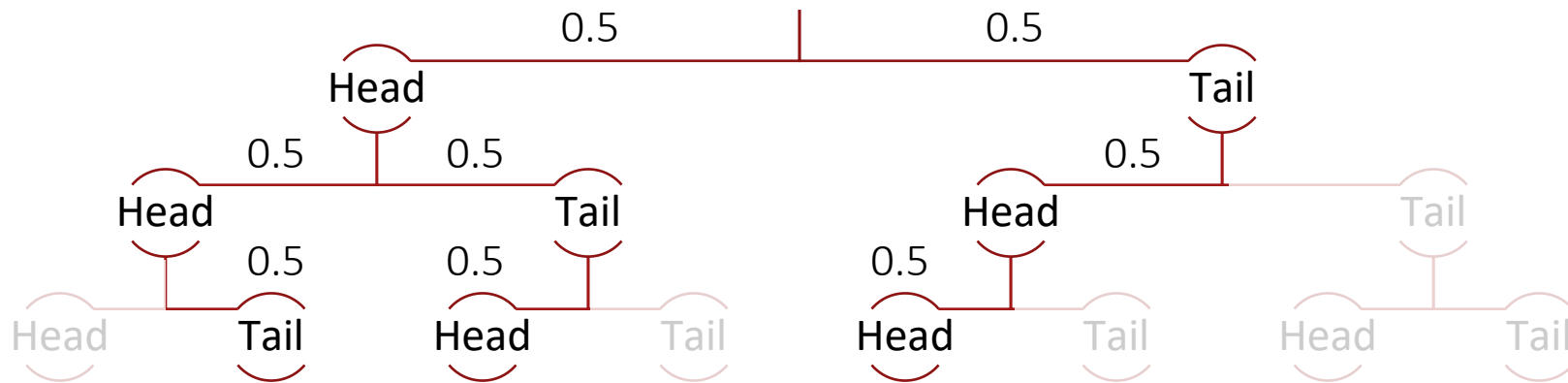
$$p^x (1 - p)^{n-x} \quad \binom{n}{x} = \frac{n!}{(n-x)!x!}$$

$$0.5^2 (1 - 0.5)^{3-2} \quad \left(\frac{3}{2}\right) = \frac{3 \times 2 \times 1}{(1)(2 \times 1)} = 3$$



The Binomial Distribution

- Used to describe binary events, by counting the occurrence of one event or the other
 - Examples?
 - Coin flip, a maybe more intuitive presentation
 - $P(2 \text{ heads})$ when tossing the coin 3 times



Probability tree

What if the coin was unfair?

What if we were playing with a dice instead?

$$\begin{aligned} &= P([H \cap H \cap T] \cup [H \cap T \cap H] \cup [T \cap H \cap H]) \\ &= [P(H) \times P(H) \times P(T)] + [P(H) \times P(T) \times P(H)] + [P(T) \times P(H) \times P(H)] \\ &= 0.5^3 + 0.5^3 + 0.5^3 \\ &= 0.375 \end{aligned}$$

What if we cared about the order?

The Binomial Distribution

- Used to describe binary events, by counting the occurrence of one event or the other

Count = (usually) discrete

Binomial distribution

Combinations

$$P(x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Failures

Successes

Parameters

n : number of trials

p : probability of success per trial

The Binomial Distribution

- Key properties of the binomial distribution
 - The outcome of a random process or experiment can result in only one of two **mutually exclusive outcomes**, such as a treatment is “success/failure”; a patient is “alive/dead” or “treated/non-treated”
 - The **outcomes are independent**, that is the outcome of any particular trial (or individual) is not affected by the outcome of any other event (or individual)
 - The **probability of a success (p) remains constant** for each trial

The Binomial Distribution

- Used to describe binary events, by counting the occurrence of one event or the other
 - Examples?
 - Likelihood of infection with rabies
 - The number of infected people (**x**) is based on the probability of successful infection after a bite (**p**) and the number of people bitten by rabid animals (**n**)

$$P(x \text{ infections}) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Modulated by
vaccine

Modulated by the
number of bites

... And the
number of rabid
animals

- Help think about mitigation measures
- Basis to compare mitigation measures



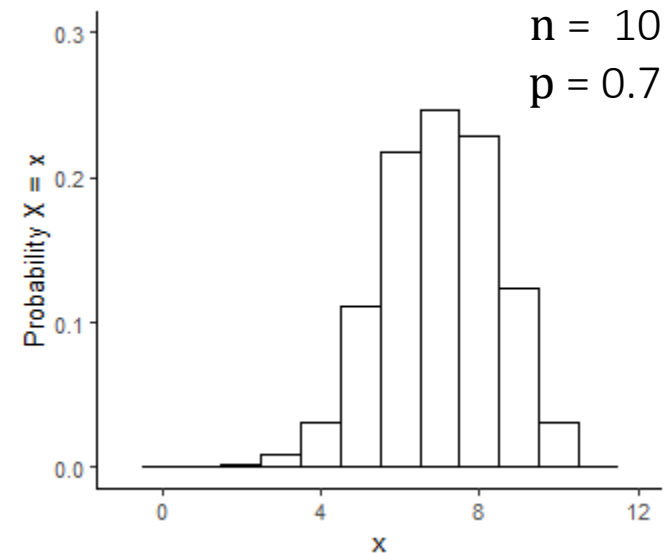
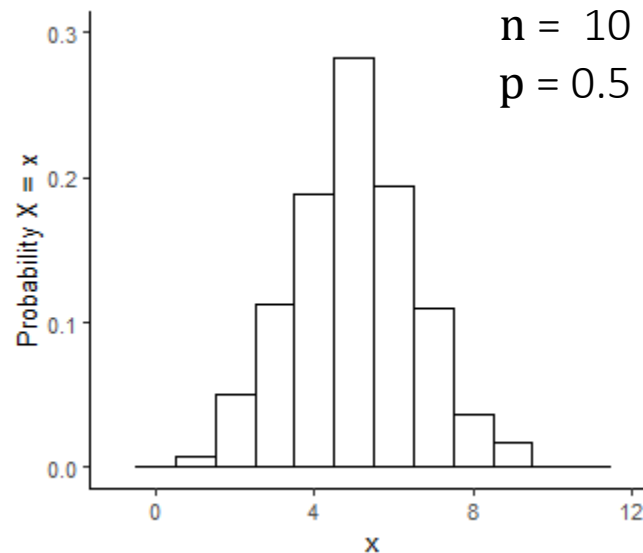
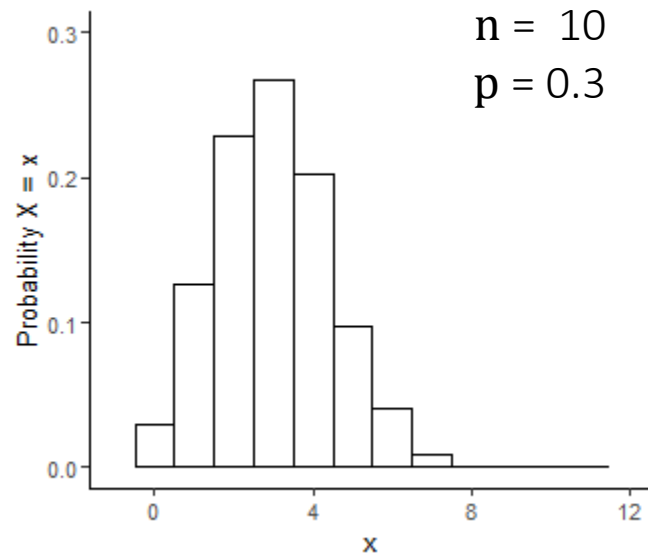
The Binomial Distribution & the Meaning of Math

- Probability mass function (PMF)

$$f(X) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- X is a discrete random variable
- Expected value $E[X] = np$

$f(X)$ is a probability $f(X) \in [0; 1]$
Variance $\text{Var}[X] = np(1 - p)$

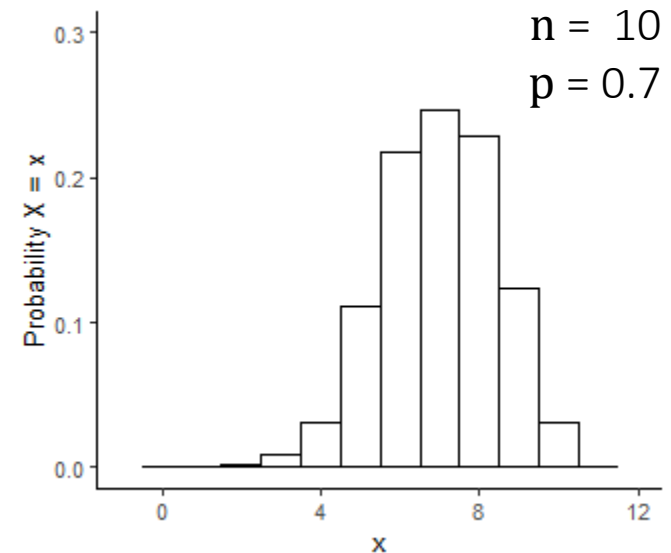
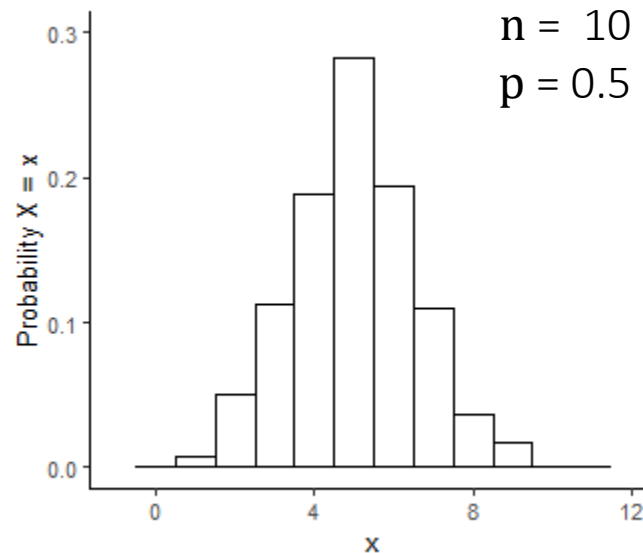
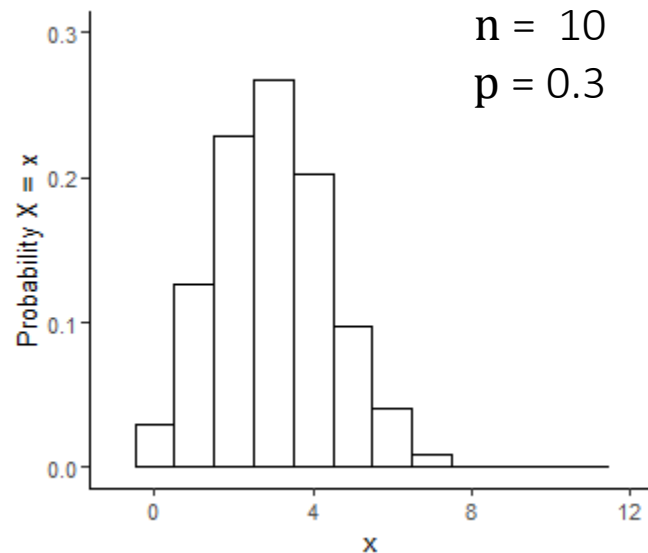


The Binomial Distribution & the Meaning of Math

- Probability mass function (PMF)

$$f(X) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- The variance is maximized for $p = 0.5$
- The binomial can be approximated with a normal distribution for very large n



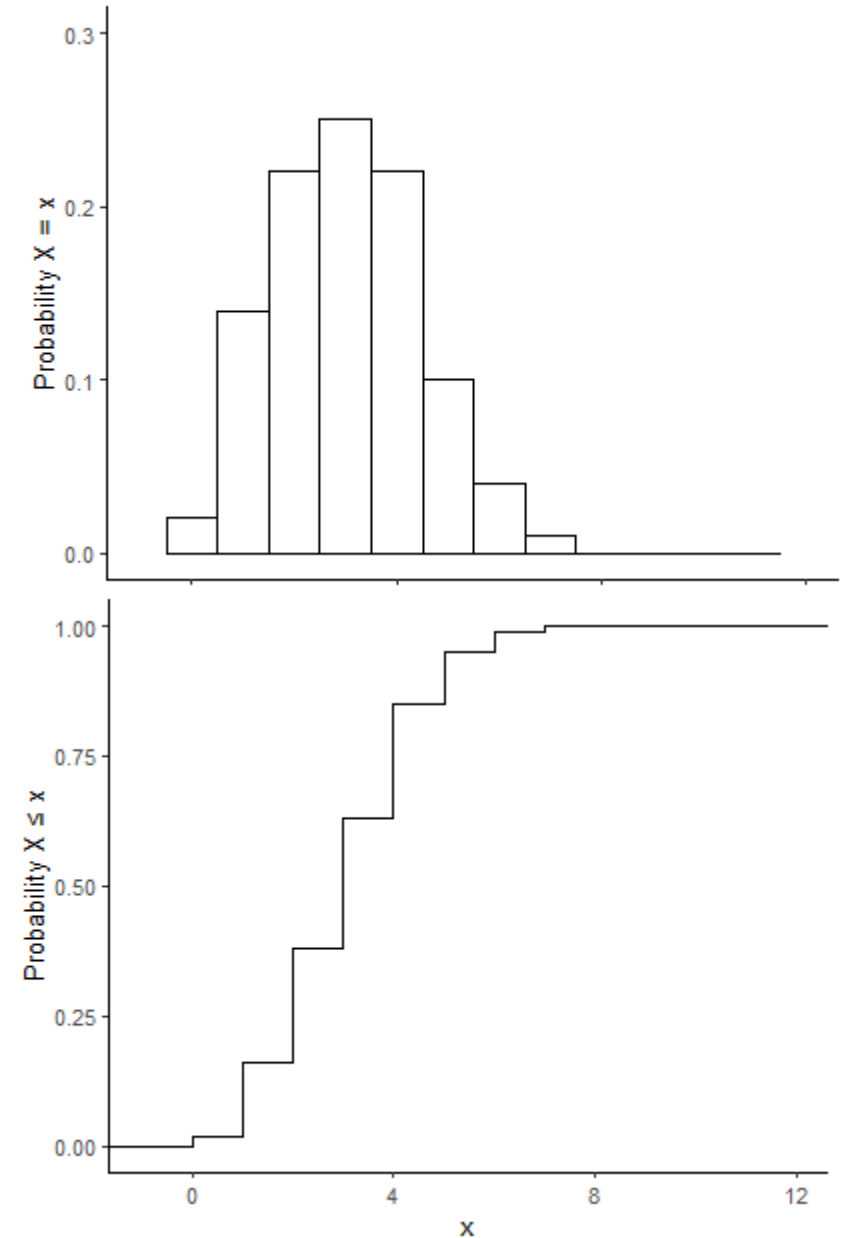
The Binomial Distribution & the Meaning of Math

- Probability mass function (PMF)

$$f(X) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- Cumulative distribution function (CMF)

$$f(X) = P(X \leq x) = \sum_{i=1}^x \binom{n}{i} p^i (1 - p)^{n-i}$$



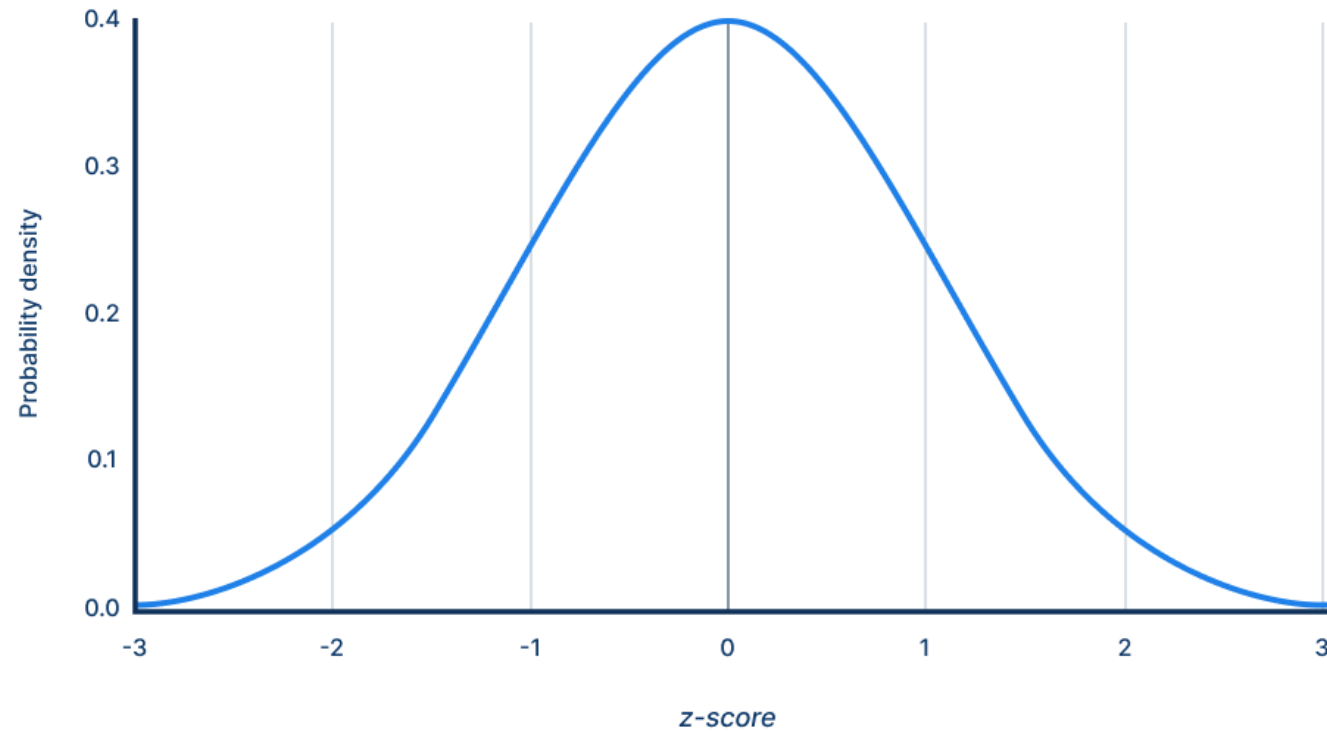
Behind the Scene: Where the Numbers Are Coming from...

The Z Distribution

Deviations from the Norm: Z Distribution and Z Score

- Z distribution = normal distribution with mean = 0 and standard deviation = 1

Standard normal distribution



The Normal Distribution, the Z Distribution, and Intervals

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$
- PDF and CDF of a normal distribution

$$N(\mu, \sigma) \quad P(\bar{x} = x) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

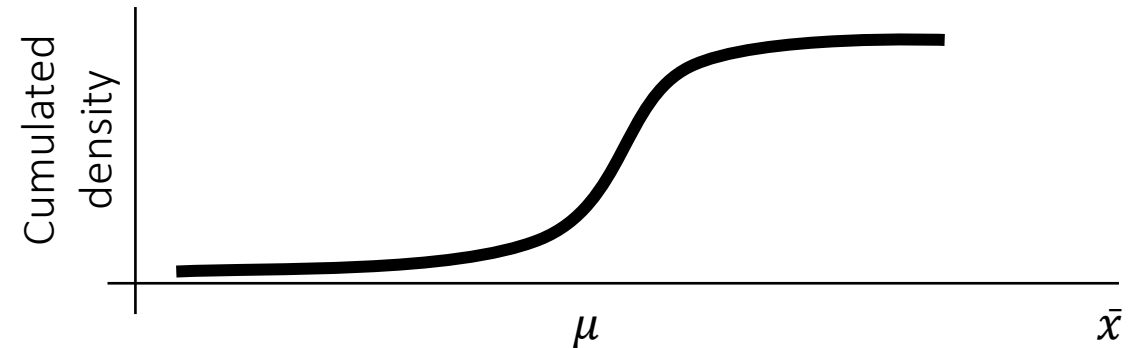
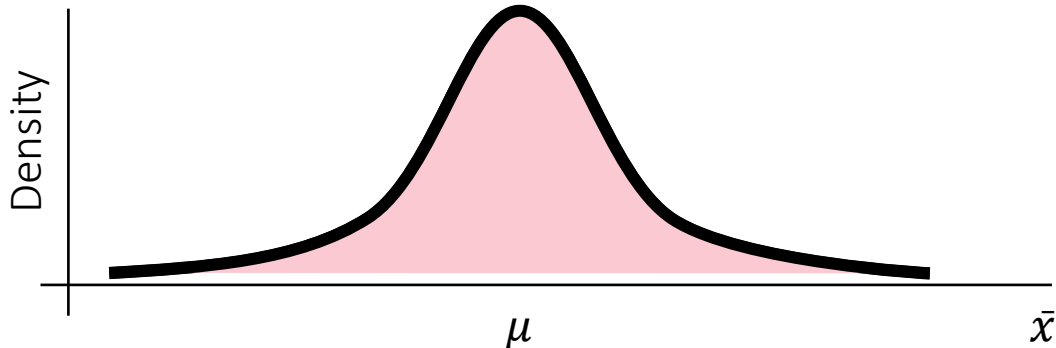
$$P(\bar{x} \leq x) = \frac{1}{\sigma\sqrt{\pi}} \int_{-\infty}^x e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Integral for all the
potential values of a
continuous variable

- PDF and CDF of the Z distribution

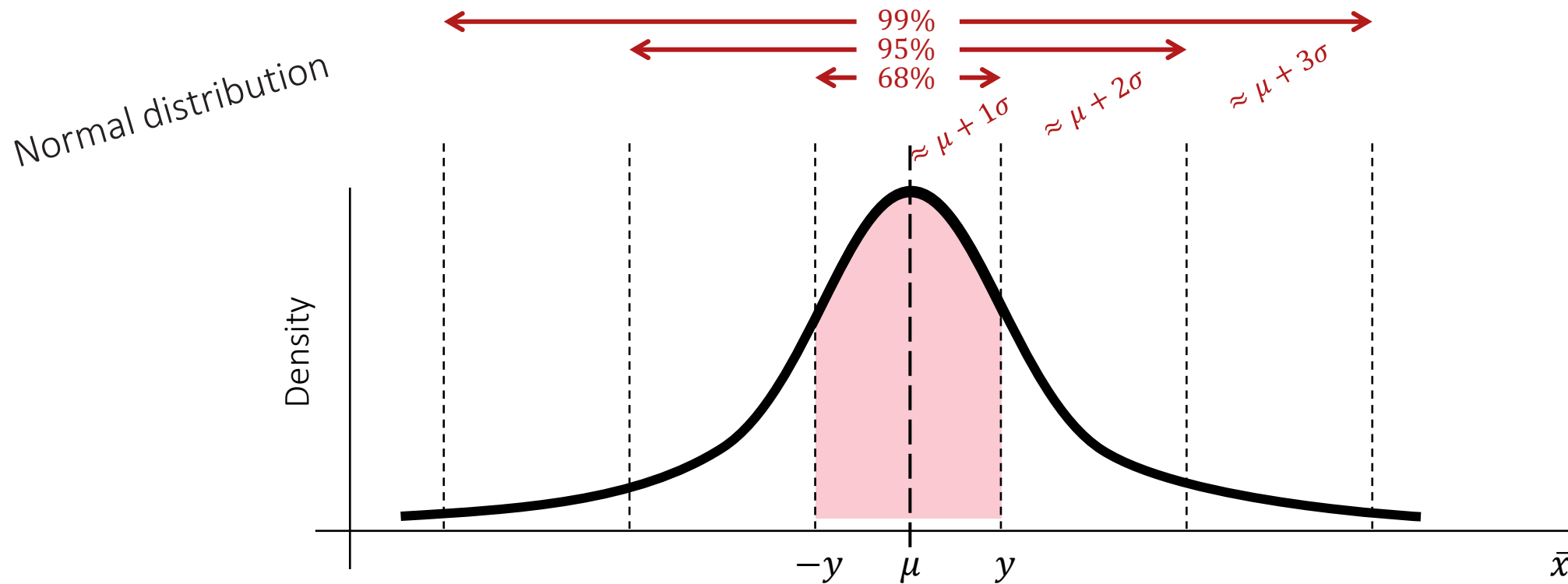
$$N(0, 1) \quad P(\bar{z} = z) = \frac{1}{\sqrt{\pi}} e^{-\frac{z^2}{2}}$$

$$P(\bar{z} \leq z) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^z e^{-\frac{z^2}{2}} dz$$



The Normal Distribution, the Z Distribution, and Intervals

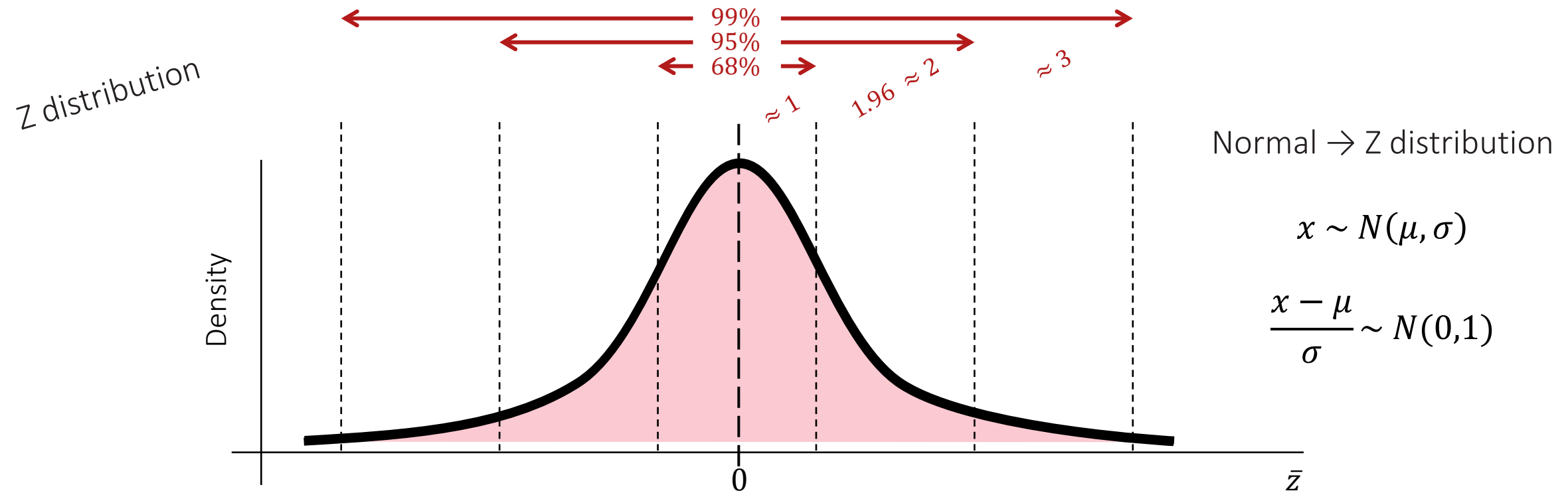
- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$



$$P(-y \leq \bar{x} \leq y) = \frac{1}{\sigma\sqrt{\pi}} \int_{-y}^y e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The Normal Distribution, the Z Distribution, and Intervals

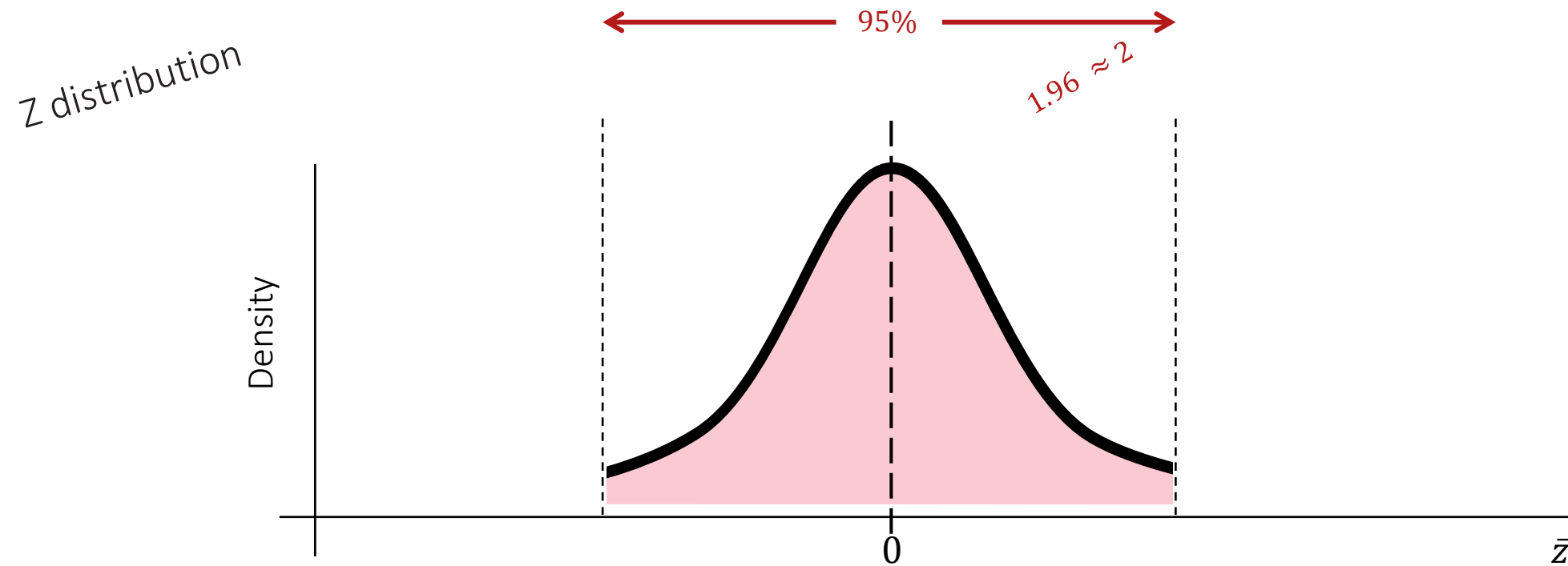
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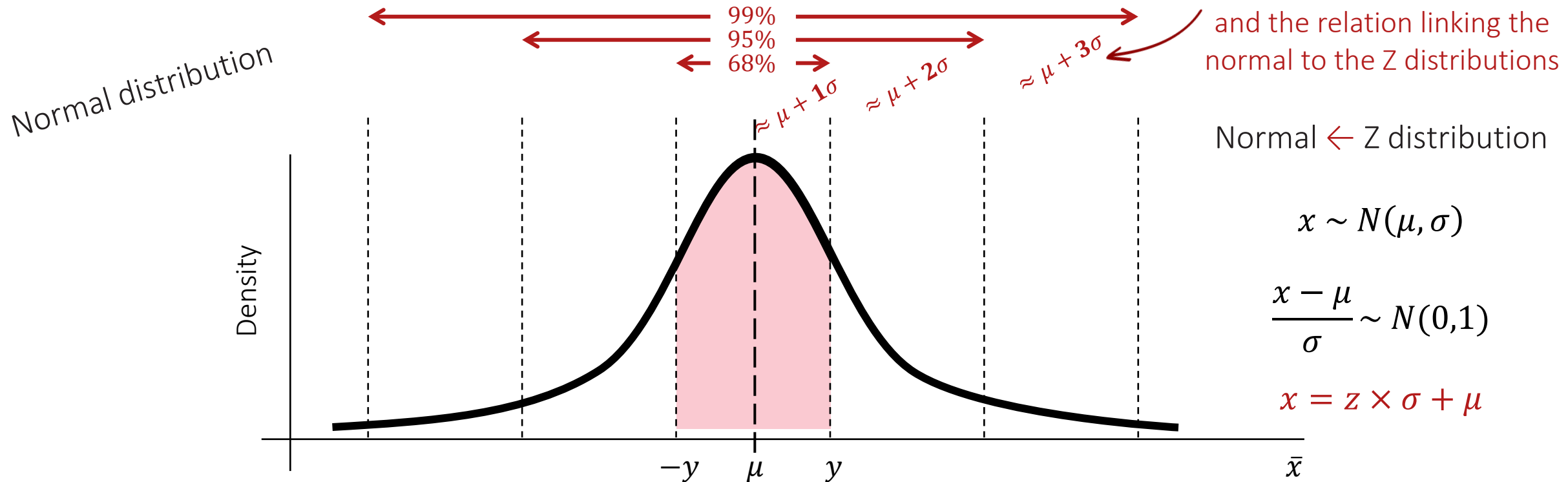


$$P(-y \leq \bar{z} \leq y) = \frac{1}{\sqrt{\pi}} \int_{-y}^y e^{-\frac{z^2}{2}} dz = 0.95 \text{ for } y = 1.96$$

This integral has no closed form, but has been tabulated

The Normal Distribution, the Z Distribution, and Intervals

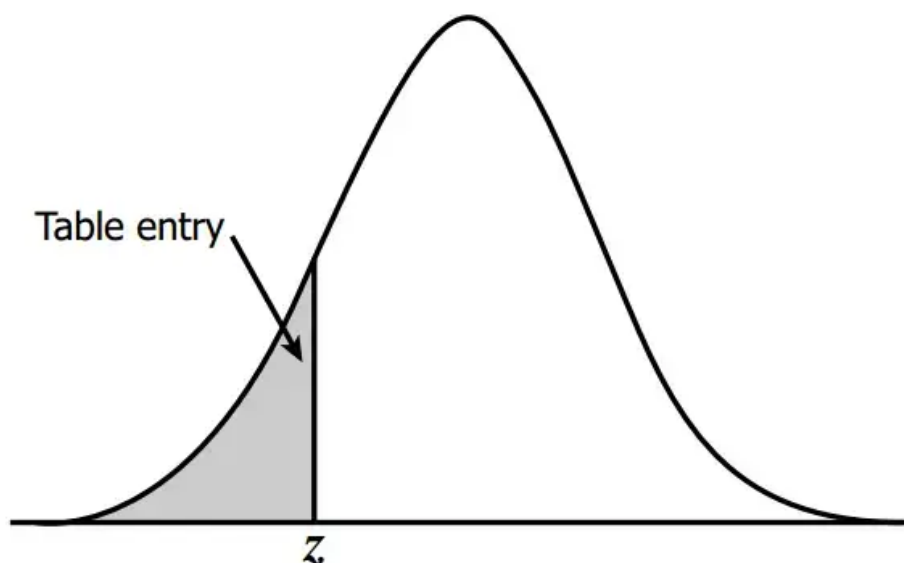
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$$P(-y \leq \bar{x} \leq y) = \frac{1}{\sigma\sqrt{\pi}} \int_{-y}^y e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The Normal Distribution, the Z Distribution, and Intervals

z-table.com



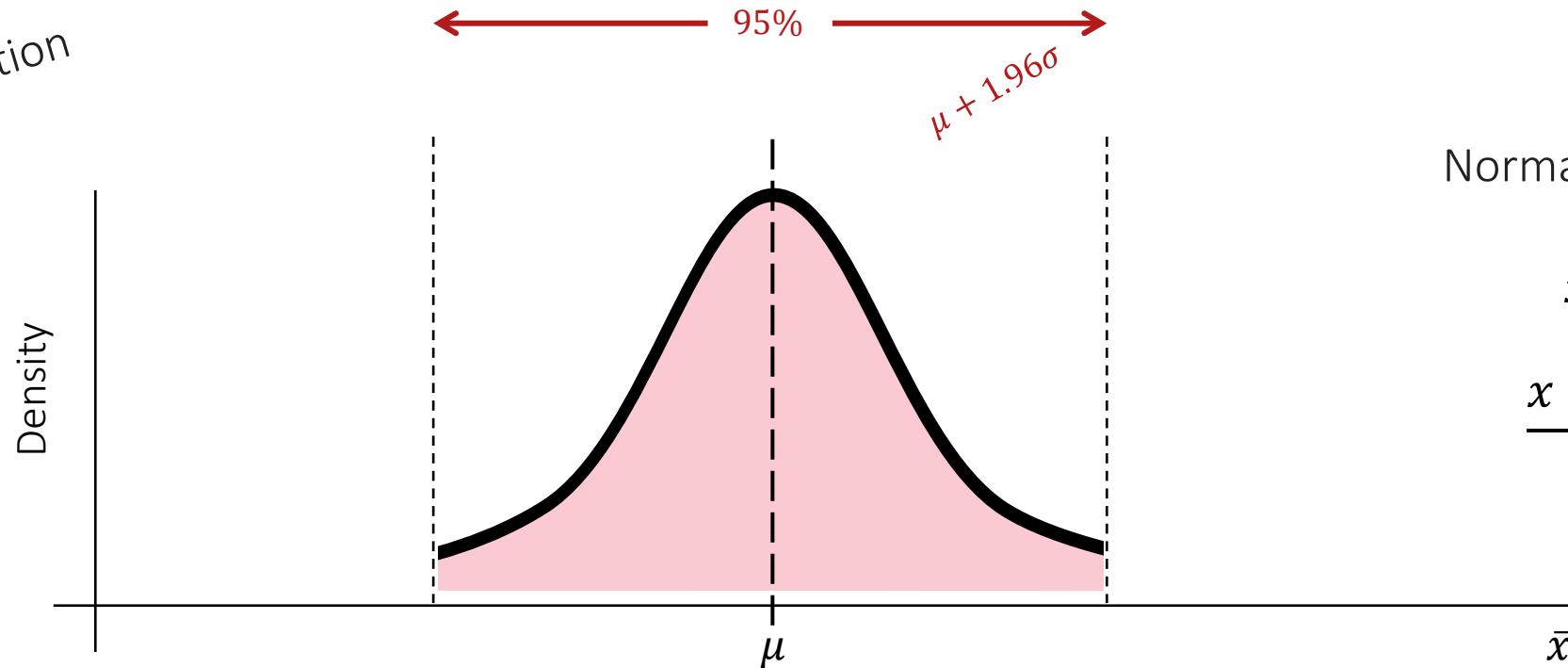
$$P(z \leq y) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^y e^{-\frac{z^2}{2}} dz$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

The Normal Distribution, the Z Distribution, and Intervals

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$

Normal distribution



Normal $\rightarrow$ Z distribution

$$x \sim N(\mu, \sigma)$$

$$\frac{x - \mu}{\sigma} \sim N(0,1)$$

Confidence Intervals: Interpretation

- The same procedure applies to means and proportions

Mean

$CI = [\bar{x} - z \frac{\sigma_s}{\sqrt{n}}; \bar{x} + z \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, n the sample size and z the z-score for the confidence level

E.g., age at death in a population

Proportion

$CI = [\bar{p} - z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}; \bar{p} + z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}]$
with $\bar{p}$ the sample proportion, n the sample size and z the z-score for the confidence level

E.g., number of infected people in a population (prevalence)

Hypothesis Testing

Testing Hypotheses...

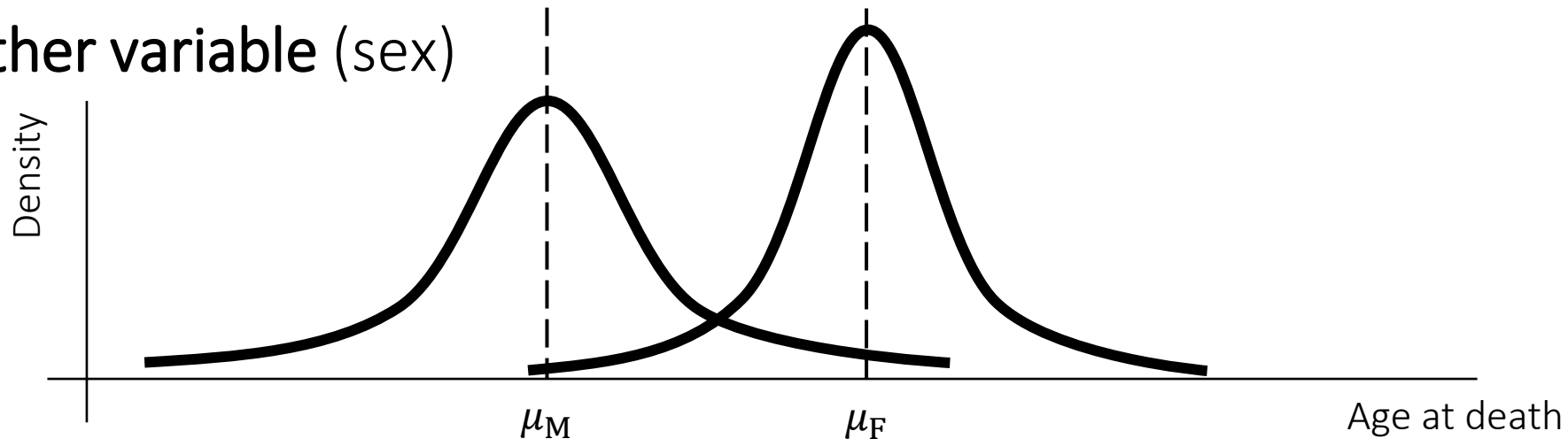
- Hypothesis testing is the foundation of modern science
- **Assessing natural factors of variations**

E.g., is sex associated with age of death?

Hypothesis: age is associated with age of death

If it is true, we expect the mean age of death for females to be different from the mean age of death for males, accounting for variations in age of death within each sex

- Notice that the **association between 2 variables** (sex and age of death) translates into a differences in the values of one variable (age) across the values of the other variable (sex)



Testing Hypotheses...

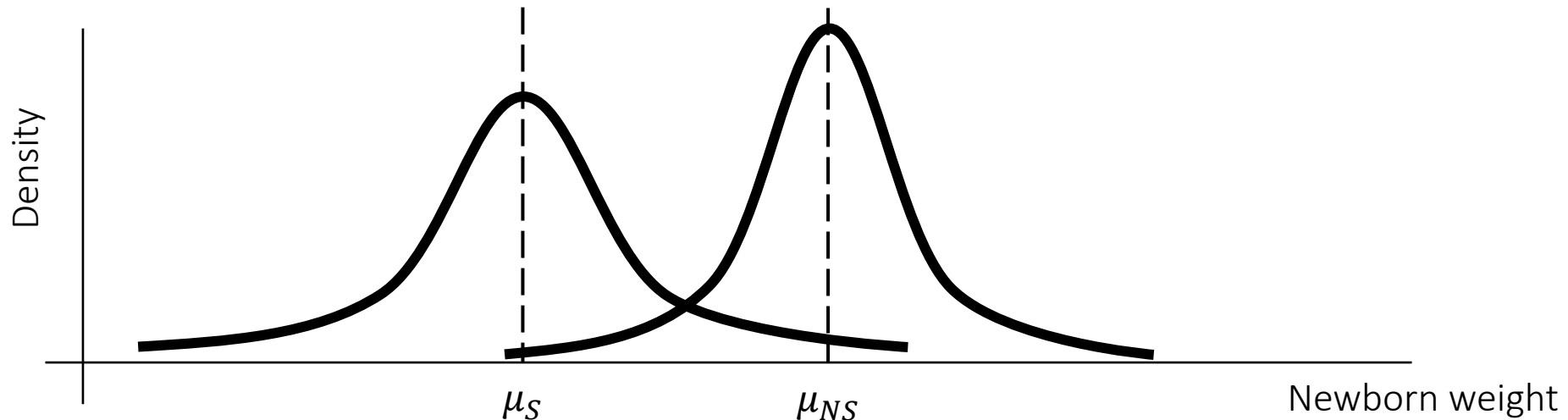
- Hypothesis testing is the foundation of modern science
- **Identifying risk factors for diseases**

E.g., is cannabis consumption associated with lower newborn weight?

Hypothesis: cannabis consumption is associated with lower newborn weight

If it is true, we expect newborns of mothers consuming cannabis to be lighter than newborns of mothers not consuming cannabis

- Notice that we can (but we don't have to) put an assumption on the **direction** of the association



Testing Hypotheses...

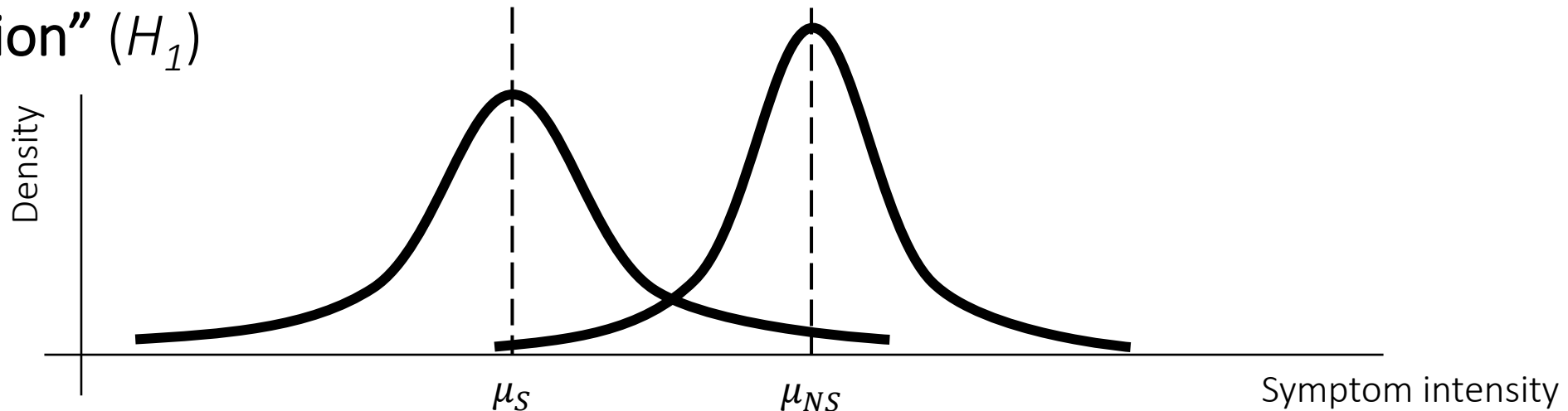
- Hypothesis testing is the foundation of modern science
- **Testing the effect of treatments**

E.g., is vitamin D supplementation associated with less severe COVID-19 symptoms?

Hypothesis: vitamin D supplementation is associated with less severe COVID-19 symptoms

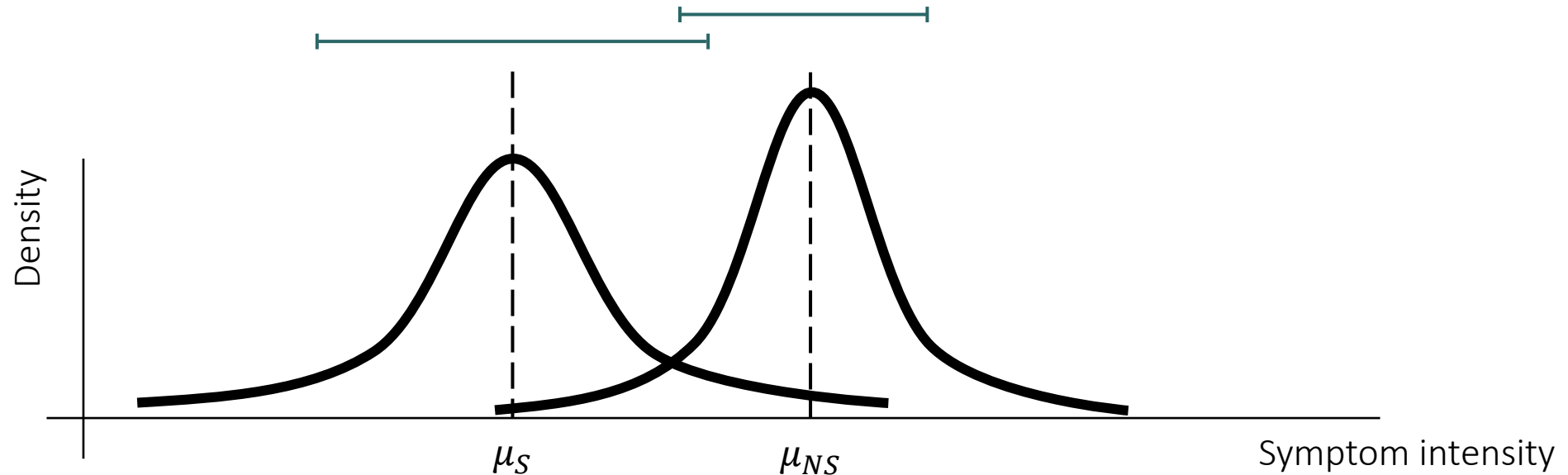
If it is true, we expect people supplemented with vitamin D to have less severe COVID-19 symptoms

- Notice that **the hypothesis is always formulated such as “there is an association”** (H_1)



Testing Hypotheses: the Concept

- Quantifying the **overlap** between the subpopulations of interest for the variable of interest, to assess whether or not the variable **significantly differs** across the subpopulations



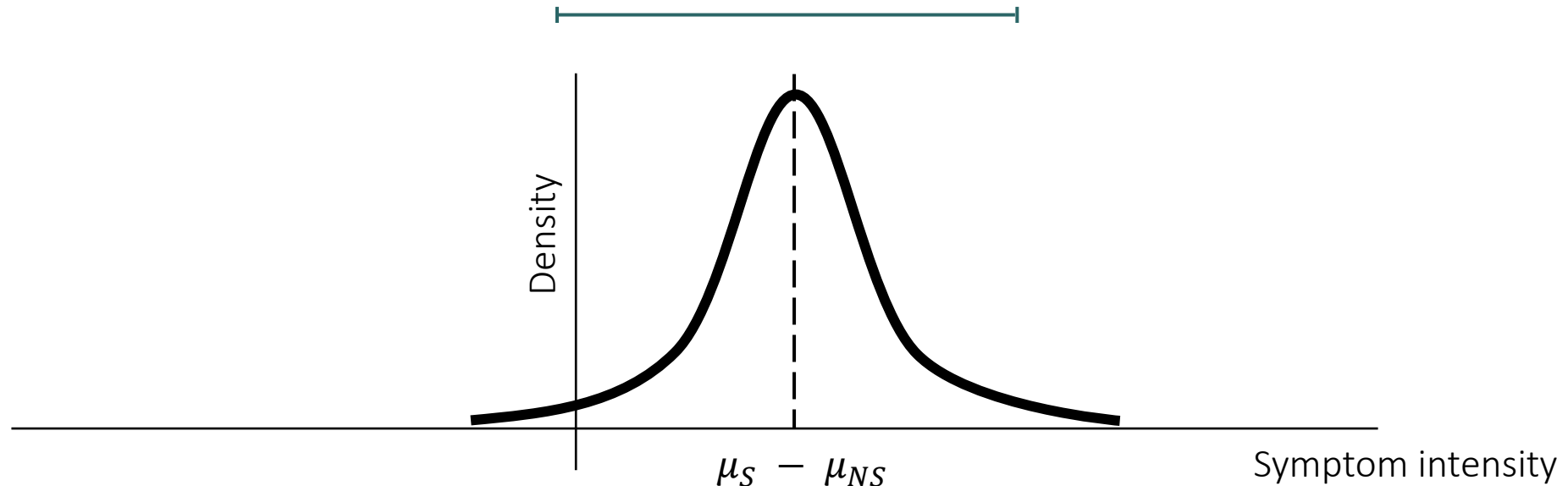
Testing Hypotheses Requires Hypotheses with Uncertainty

- Formulate a hypothesis, corresponding to the absence of association (H_0)
- Test it
- Reject it or not, with a given level of confidence (α)

Null hypothesis is...	True ($\mu_1 = \mu_2$)	False ($\mu_1 \neq \mu_2$)
Reject	Type I Error False positive Probability = α	Correct decision True positive (association) Probability = $1 - \beta$
Fail to reject	Correct decision True negative (no association) Probability = $1 - \alpha$	Type II Error False negative Probability = β

Testing Hypotheses: the Concept

- Quantifying the **overlap** between the subpopulations of interest for the variable of interest, to assess whether or not the variable **significantly differs** across the subpopulations
- Another way to do it: **quantifying the overlap with 0 of the difference of the variable between the subpopulations**



Testing Hypotheses Requires Hypotheses with Uncertainty

- Formulate a hypothesis, corresponding to the absence of association (H_0)
- Test it
- Reject it or not, with a given level of confidence (α)

Null hypothesis is...	True ($\mu_1 = \mu_2$) $\mu_1 - \mu_2 = 0$	False ($\mu_1 \neq \mu_2$) $\mu_1 - \mu_2 \neq 0$
Reject	Type I Error False positive Probability = α	Correct decision True positive (association) Probability = $1 - \beta$
Fail to reject	Correct decision True negative (no association) Probability = $1 - \alpha$	Type II Error False negative Probability = β

Confidence Intervals for Differences

Mean

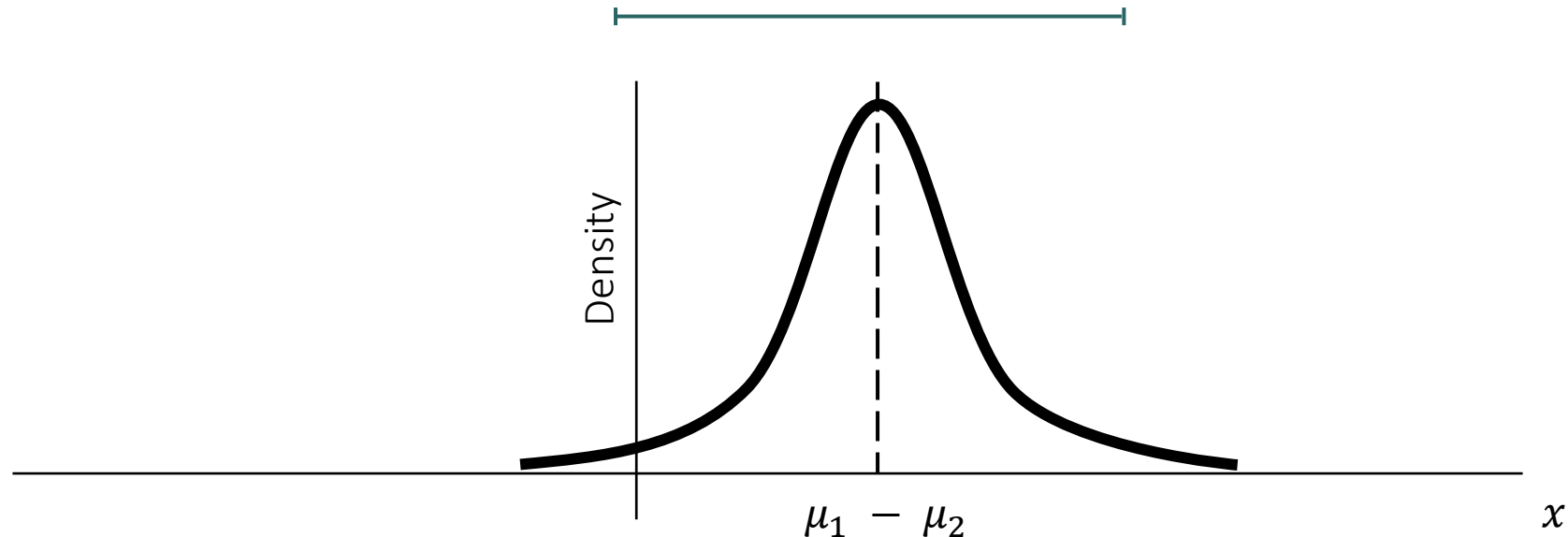
$$CI = (\bar{x}_1 - \bar{x}_2) \pm z \sqrt{\frac{\sigma_{s1}^2}{n_1} + \frac{\sigma_{s2}^2}{n_2}}$$

Proportion

$$CI = (\bar{p}_1 - \bar{p}_2) \pm z \sqrt{\frac{\bar{p}_1(1-\bar{p}_1)}{n_1} + \frac{\bar{p}_2(1-\bar{p}_2)}{n_2}}$$

If the variance is the same

$$CI = (\bar{x}_1 - \bar{x}_2) \pm z \sigma_s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$



Confidence Intervals for Differences

Mean

$$CI = (\bar{x}_1 - \bar{x}_2) \pm z \sqrt{\frac{\sigma_{s1}^2}{n_1} + \frac{\sigma_{s2}^2}{n_2}}$$

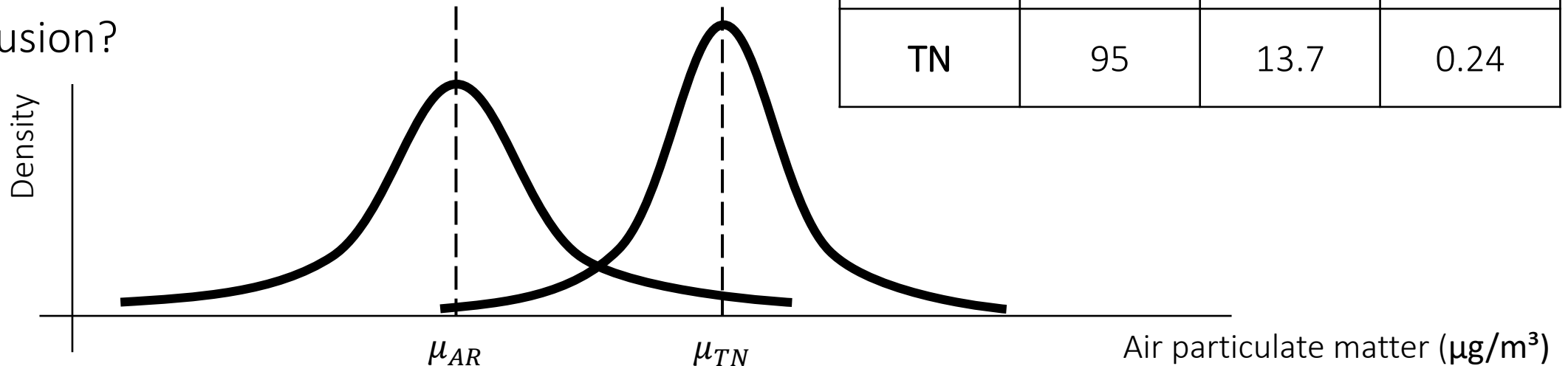
$$CI = -2.16, -1.84$$

We reject the hypothesis of no difference with confidence level 95%

- Compare the average pollution by air particulate matter between Arkansas and Tennessee

- H_0 and H_1 ?
- CI of the difference?
- Conclusion?

<i>Data</i>	<i>n</i>	$\bar{x}$	σ_s
AR	75	11.7	0.32
TN	95	13.7	0.24



Take Home Messages

Always report a central
tendencies with measures
of dispersion

Pick the right summary
statistics for your data (based
on their distribution)

Always report a point
estimate with measures of
uncertainty

Always accompany summary
statistics with some
visualization of the raw data

Increasing sample size
enables a higher precision
with high confidence
i.e., lower uncertainty)

We never know the true
population mean, but we know
the distribution of the sample
mean is centered around it, so
we can infer where it likely is

Before Next Time (Friday February 13, in LH4)

- Practice: [Homework #03](#) due by Thursday 5 PM
- Use the [Guiding Quiz #03](#) if needed
- Check you are ready for next class: [Knowledge Check #03](#) due by Friday 11 AM

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
- Come to office hours
 - **Tuesdays, 09:00-10:00 AM** (Jordan, Maddy)
 - **Wednesdays, 05:00-06:00 PM** (Mainul, Princal, Daniel and Harsh)
 - If you cannot make it at these times, please contact the course assistant of your choice by email
 - In S1-122, unless indicated otherwise on Canvas